



MSc Data Analytics

Final Year Project

Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies

By

Asad Ikram (UP2291687)

September 2025

Project Unit: U22244 Masters

Project Supervisor: Dr Ella Haig

Computing and Mathematics Program Area

Abstract

Public sentiment on economic policies often first appears on social media, spread across platforms such as Twitter (X), Reddit, YouTube, and Quora. This dissertation presents a cross-platform sentiment analysis framework designed to unify and forecast citizen reactions to UK economic policies. Approximately 279,000 public social-media posts and comments related to taxation, welfare, inflation, and stimulus measures were collected using official APIs and licensed feeds. Posts were cleaned and labeled using the VADER sentiment analysis tool, and further analyzed using Natural Language Processing (NLP) and other techniques. A transformer-based model (BERT) and a Bidirectional LSTM were trained to classify sentiment (positive, neutral, negative), achieving high accuracy (BERT macro-F1 ≈ 0.88) on a validated set. Sentiment trends were aggregated over time, and Facebook Prophet was applied to forecast short-term public mood. The results reveal notable sentiment differences across policy domains, while the forecast model captured seasonal patterns and reactions around major policy announcements. Ethical and GDPR-compliant data handling was ensured, graphs were developed for policy analysts. The findings demonstrate that cross-platform NLP can effectively gauge public opinion in real time, providing policymakers with a proactive tool to understand and anticipate public response.

Keywords: UK economic policy, sentiment analysis, social media, BERT, BiLSTM, NRC, Prophet, calibration, explainability.

Word count: 12,626 words

Acknowledgements

I would like to thank my supervisor, Dr. Ella Haig, for her invaluable guidance and feedback throughout this project. I am also grateful to the University of Portsmouth School of Computing for providing the resources and support needed to carry out the research. Special thanks to my friends and colleagues in the MSc Data Analytics program for their encouragement and for the insightful discussions that helped shape this work.

Table of Contents

Chapter 1. Introduction	6
Chapter 2. Literature Review.....	8
2.1 Search Strategy and Source Credibility	8
2.2 Sentiment Analysis of Government Policies on Social Media	8
2.3 Gaps and Research Opportunities	10
Chapter 3. Project Management and Professional Issues	12
3.1 Timeline and Milestones	12
3.2 Resource Management	12
3.3 Risk Management	13
3.4 Deviation Log and Lessons Learned	14
3.5 Data Protection and Privacy (GDPR)	14
3.6 Legal Compliance	15
3.7 Ethics and Social Responsibility	15
Chapter 4. Specification & Discussion of the Research Goals	16
4.1 Primary Goals	16
4.2 Optional Goals	19
Chapter 5. Data Collection	21
5.1 Sources and Tools	21
5.2 Data Storage and Governance	25
5.3 Data Quality and Cleaning Verification	25
5.4 Sentiment Label Verification:	25
5.5 Limitations of Data Collection.....	25
5.6 Summary	25
Chapter 6. Methodology	26
6.1 Research Design and Rationale	26
6.2 Data Preprocessing Pipeline	27
6.3 Sentiment Classification Models	29
6.4 Emotion Analysis	30
6.5 Time-Series Forecasting.....	30
6.6 Evaluation Plan.....	32
6.6 Implementation and Reproducibility.....	34

Chapter 7. Results Analysis & Discussion	35
7.1 Sentiment Classification Performance	35
7.2 Cross-Platform Sentiment Trends and Forecasting	38
7.3 Diagnostics, Interpretability, and Policy-Level Variation.....	40
7.3 Model Agreement, Disagreement, and Significance	42
7.4 Summary of Findings	44
Chapter 8. Evaluation Against Research Goals	46
8.1 Goal 1: Multi-platform data aggregation – Achieved.....	46
8.2 Goal 2: Sentiment classifier accuracy $\geq 70\%$ macro-F1 – Exceeded.....	46
8.3 Goal 3: Temporal trends visualization – Achieved	47
8.4 Goal 4: Short-term sentiment forecasting – Achieved	47
8.5 Goal 5: Policy Topic Categorization – Achieved	47
8.6 Goal 6: Explainable Model Outputs – Mostly Achieved.....	48
8.7 Additional Goals	48
8.8 Evaluation Summary.....	48
8.9 Limitations in Evaluation.....	48
8.10 Strengths and Limitations Reflection	49
Chapter 9. Conclusion.....	50
9.1 Summary of Findings	50
9.2 Implications for Potential Stakeholders.....	50
9.3 Limitations and Future Work	51
9.4 Reflections on Methodology.....	52
9.5 Recommendations and Future Outlook.....	52
9.6 Overall Conclusion	52
Chapter 10. References	53
10.1 References	53
10.2 Bibliography.....	54
Chapter 11. Appendices	55
11.1 Appendix A Project Specification	55
11.2 Certificate of Ethics Review	60

Chapter 1. Introduction

Problem

Public reaction to government economic policies such as tax changes, welfare reforms, inflation control measures, and stimulus packages now emerges rapidly on social media. Traditionally, policymakers have relied on opinion polls and single-platform analytics to gauge public opinion; however, these methods can be slow and fragmented. With citizen discussions dispersed across Twitter (recently rebranded as X), Reddit forums, YouTube comment sections, Quora threads, and other venues, assembling a coherent and timely picture of sentiment presents a significant challenge. Important early warning signs of public discontent may be missed if monitoring is confined to a single platform or updated infrequently. This project addresses this gap by developing a cross-platform sentiment analysis framework capable of capturing and making use of public sentiment across multiple social media sources in real time.

Aim and Objectives

The primary aim of this dissertation is to develop a research-grade workflow that mines social-media data for insights into UK public sentiment on economic policy and to evaluate whether machine learning techniques can effectively predict future sentiment trends. Four key objectives have been defined:

1. **Data Ingestion & Integration:** Secure ingestion of UK-focused public posts and comments from Twitter, Reddit, YouTube, and Quora via official APIs or licensed data feeds, consolidated into a unified dataset.
2. **Sentiment Classification:** Cleaning and preprocessing of text data, followed by sentiment labeling using the VADER sentiment analysis tool. Further classification of sentiment (positive, negative, neutral) with respect to specific economic-policy topics was carried out using state-of-the-art machine learning models.
3. **Trend Analysis:** Aggregation of sentiment scores to identify temporal trends and cross-platform similarities or differences in public reaction.
4. **Predictive Forecasting:** Generation of short-term sentiment forecasts (e.g., weeks ahead) using time-series models to assess whether past trends can predict future mood, particularly around major policy announcements.

Research Questions:

In line with the aims and identified gaps in the literature (see Section 2), the study addresses the following research questions (RQs):

- RQ1: How effective are modern NLP sentiment analysis methods (e.g., transformer models and lexicon-based approaches) at capturing public sentiment towards UK economic policies on social media?
- RQ2: How accurately can transformer-based and BiLSTM models classify posts as positive, neutral, or negative on held-out validation data?

- RQ3: Can temporal patterns in sentiment be leveraged to reliably forecast short-term public reactions to upcoming economic policy decisions?
- RQ4: How do sentiment dynamics vary across key dimensions such as policy topic, time period within the UK?

Scope and Assumptions:

The study focuses on major UK economic policy discussions from approximately 2010 to mid-2025, including topics such as tax policy changes (e.g., budget announcements), welfare program reforms, inflation and interest rate measures, and government stimulus efforts. Only public posts in English are analyzed, under the assumption that these reflect a substantial portion of UK public discourse online. Analysis is restricted to the four selected platforms due to data availability, depth of discussion and API access; other networks such as Facebook or Instagram are excluded. While social media users are not fully representative of the population, prior research suggests that these platforms provide meaningful insights into public opinion trends (see Section 2). All data used were publicly available and accessed in compliance with platform policies and UK data protection laws. No personal identifying information was retained, and results are reported in aggregate to ensure GDPR compliance (see Section 5).

Contributions:

This dissertation makes three primary contributions. First, it develops an integrated cross-platform data pipeline that, to our knowledge, is among the first to simultaneously analyse multiple social media sources for UK policy sentiment in a single framework publicly available. Second, it evaluates advanced machine learning models (a fine-tuned BERT transformer and a BiLSTM network) on a novel aggregated dataset of economic policy discussions, demonstrating high accuracy in sentiment classification and highlighting the strengths of transformer-based approaches. Third, it introduces a preliminary sentiment forecasting module using time-series modelling (Prophet), providing initial evidence that social media sentiment trends can offer predictive signals about shifts in public mood. Practical outputs include graphs for policy analysts to interact with the data and detailed documentation to support reproducibility and responsible model use. By addressing the stated research questions, the study demonstrates that real-time, cross-platform sentiment analysis can enable government agencies to communicate more effectively and adjust policies proactively in response to citizen feedback.

Chapter 2. Literature Review

2.1 Search Strategy and Source Credibility

To situate this research, a comprehensive review of current literature was conducted, focusing on social media sentiment analysis in government and policy contexts. Literature searches were performed using academic databases including IEEE Xplore, ResearchGate, ACM Digital Library, SpringerLink, and Google Scholar. Keywords included combinations of “sentiment analysis”, “government policy”, “social media”, “public opinion”, “Twitter”, “Reddit”, “cross-platform”, and specific policy-related terms (e.g., “tax sentiment analysis”). The search targeted publications from approximately 2015 through 2025 to capture recent advances particularly the emergence of deep learning methods while incorporating influential earlier works in sentiment analysis.

Approximately 30 relevant sources were identified, from which a subset of highly pertinent studies was selected for critical review. Inclusion criteria emphasized peer-reviewed articles, reputable conference proceedings, and institutional reports focusing on public sentiment or opinion mining in policy/government domains. Source credibility was evaluated based on the publisher (with preference given to established journals and conferences), author expertise, citation metrics, and, where applicable, transparency in methodology and data usage. Greater weight was assigned to studies employing robust data collection and validation procedures.

This search strategy ensured that the literature review encompassed credible and relevant findings at the intersection of Natural Language Processing (NLP), social media, and policy sentiment.

2.2 Sentiment Analysis of Government Policies on Social Media

In recent years, growing interest has been observed in the use of social media data to gauge public reaction to government decisions. Sentiment analysis has been applied in various political and policy contexts across the globe. For instance, a 2023 study in the Philippines by Vizmanos et al. examined Facebook reactions to development policies by applying opinion mining to comments from three viral posts published by a government think tank (Albert et al., 2023). Comment sentiment was categorized as positive, neutral, or negative, and the findings demonstrated that social media feedback could provide real-time insights that complemented traditional surveys. The study was primarily descriptive, focusing on current sentiment towards policies such as free college tuition and a cash transfer program, and highlighted how public Facebook comments can reveal challenges faced by citizens.

Similarly, research in Italy (Corallo et al., 2015) analysed tweets concerning a cultural event Lecce’s bid for the 2019 European Capital of Culture as a proxy for government-related sentiment. Classical machine learning classifiers, including Naïve Bayes and Support Vector Machines (SVM), were applied alongside a dataset-level sentiment estimation algorithm (ReadMe). After part-of-speech tagging and the use of n-gram features, Naïve Bayes achieved approximately 78% classification accuracy. These results indicated that even traditional models, when appropriately tuned and trained on sufficient data, could effectively classify public sentiment on civic topics. However, the study was limited to a single event and platform

(Twitter), with sentiment aggregation conducted only at the document or dataset level, lacking a broader cross-platform or longitudinal analysis.

Several studies in India have utilized Twitter data to analyse sentiment on government decisions, demonstrating the feasibility of mining public opinion on policy reforms. It (Anand et al., 2020) presented a survey of sentiment analysis techniques applied to major Indian government decisions such as the Goods and Services Tax, a citizenship act, and demonetization. Using supervised classifiers, including Naïve Bayes and logistic regression, alongside lexicon-based methods, tweets were classified as positive, negative, or neutral to observe public opinion trends. The authors reported that a substantial fraction of tweets remained neutral, reflecting reluctance or caution among citizens when discussing controversial decisions.

A related study by another individual (Rekha et al., 2019) focused on sentiment towards specific Indian economic schemes, including the 2016 demonetization policy and the GST tax reform, using data collected via the Twitter API. Machine learning classifiers such as Random Forest, SVM, and a tailored Random Forest were applied to a corpus of filtered tweets containing relevant hashtags. The models achieved approximately 75–80% classification accuracy with reasonably strong precision and recall, demonstrating that public tweets can be mined effectively for policy feedback. The study found that sentiment on Twitter was predominantly negative in the immediate aftermath of demonetization, aligning with public discomfort reported in conventional media. However, the approach was limited to a single platform and retrospective in nature, examining only historical data without predictive modelling.

Beyond sentiment classification, some research has sought to correlate sentiment with actual policy outcomes or changes. It (Srivastava et al., 2020) investigated public sentiment surrounding India's demonetization reform by collecting approximately 1,000 tweets at regular intervals from November 2016 (policy introduction) through March 2017. Sentiment polarity was determined by counting positive and negative words in tweets, after which Pearson correlation coefficients were computed between the expected sentiment (based on the government's anticipated public reception) and observed sentiment trends. A moderate positive correlation ($r \approx 0.65$) was found for positive sentiment, and a similarly moderate negative correlation ($r \approx -0.67$) was found for negative sentiment. These findings suggest that public sentiment partially tracked policy adjustments over time. This study is one of the few to attempt quantification of the relationship between policy actions and social media sentiment, effectively treating sentiment as a real-time barometer of policy acceptance. Although restricted to Twitter and one country, the results reinforce the value of sentiment monitoring during policy rollout and indicate potential for forecasting sentiment in response to future reforms.

Related domains also provide methodological insights into sentiment analysis. They (Souza et al., 2024) applied transformer-based sentiment analysis to tweets concerning Brazilian National Parks to inform environmental policy. Using a BERT model fine-tuned for Portuguese (BERTimbau), overall accuracy of approximately 81% was achieved. Class-wise performance revealed strong detection of negative sentiment ($F1 \approx 0.83$) but weaker performance for neutral

($F1 \approx 0.23$) and positive ($F1 \approx 0.44$) classes, attributed to class imbalance and the subtlety of positive expressions in the data. This highlights a common challenge in sentiment analysis, where models detect explicit negative expressions more reliably than nuanced neutral or mildly positive content. The Brazilian study also incorporated topic modeling (BERTopic) to identify thematic concerns such as wildlife and park management underlying sentiment. Although the context differed from economic policy, the methodology illustrates that transformer models can deliver strong classification performance, and that combining sentiment with topic analysis can enrich the interpretation of underlying public attitudes.

Fine-grained opinion monitoring has also been applied in public service contexts. It (Chen et al., 2018) analysed public comments on suburban parks in Shanghai from platforms such as Weibo and Dianping to assess government service quality. Reviews were segmented, scored using a sentiment dictionary specific to Chinese, and evaluated via regression analysis to determine correlations between sentiment scores for specific aspects (e.g., environment, facilities) and official satisfaction ratings. Positive correlations were observed for attributes such as transportation convenience and food availability with overall satisfaction, validating those online sentiments aligned with on-ground service quality perceptions. While this study did not address economic policy directly, it demonstrates the application of aspect-based sentiment analysis in the public sector. Its reliance on a manual sentiment lexicon and simple positive/negative word counts was a limitation, as such approaches may fail to capture sarcasm or nuanced context. Nevertheless, it underscores the growing practice of governments leveraging social media sentiment to monitor and improve public services in real time.

Collectively, these works motivate combining robust classifiers with lexicons for interpretability (Hutto & Gilbert, 2014; Mohammad & Turney, 2013) and leveraging transformer models where domain drift and context sensitivity matter (Devlin et al., 2019).

2.3 Gaps and Research Opportunities

The reviewed literature collectively demonstrates that sentiment analysis has been applied to understand public opinion on government actions in various contexts, ranging from social welfare to demonetization to cultural events. However, several notable gaps remain.

Firstly, cross-platform analyses are rare, with most studies focusing on a single platform typically Twitter or Facebook in isolation (Albert et al., 2023; Corallo et al., 2015; Rekha et al., 2019). This presents a significant limitation, as different demographics use different platforms; a sentiment signal observed on Twitter may differ in magnitude or timing from that on YouTube or Reddit. The present research addresses this limitation by comparing sentiment for the same set of issues across multiple platforms simultaneously, enabling a more comprehensive view of public opinion.

Secondly, most prior studies have been descriptive or retrospective, analyzing historical data to explain past trends, with relatively few attempts to predict or forecast sentiment trajectories. Given the moderate correlations found in studies such as Srivastava's (Srivastava et al., 2020) work on demonetization, there is a clear opportunity to test whether incorporating recent sentiment patterns can be used to predict near-future public mood. Incorporating decomposed forecasting can add value in policy settings (Taylor & Letham, 2018).

Thirdly, granular breakdowns by topic and time have often been limited. Many studies aggregate sentiment across an entire event or policy, whereas this research examines sentiment disaggregated by specific economic policy categories (e.g., tax, inflation, welfare) over a multi-year timeline encompassing multiple policy events. This enables observation of patterns such as whether certain policy types consistently trigger stronger negative sentiment or whether sentiment improves over time as policy impacts unfold.

Fourthly, another identified gap lies in the interpretability and trustworthiness of sentiment models. Many reviewed studies (Ribeiro et al., 2016; Hutto & Gilbert, 2014; Mohammad & Turney, 2013) report accuracy metrics but provide limited explanation of the features or linguistic cues driving model decisions, with exceptions such as Chen's (Chen et al., 2018) use of a known sentiment dictionary. Given that policy analysts require trust in the insights produced, this research incorporates interpretability techniques such as LIME explanations and lexicon cross-checks to verify that model outputs align with logical indicators (e.g., anger-related terms contributing to negative classifications). Additionally, challenges such as sarcasm, idiomatic expressions, and coordinated political messaging are frequently acknowledged but rarely examined in depth. Using advanced transformer-based models (BERT) and systematic error analysis, this study highlights such challenges and proposes directions for future work.

In summary, the literature indicates that social media sentiment analysis is a promising tool for understanding public opinion on government policies, yet the field remains largely undiscovered by platform and underdeveloped in its predictive capacity. No prior study was identified that integrates multiple social media sources for the analysis of UK economic policy sentiment while also attempting to forecast sentiment change. This research addresses that gap by building on established methods such as machine learning classifiers, lexicon-based approaches, and correlation analysis. Extending them into a unified multi-platform framework incorporating predictive analytics. Lessons from previous studies have informed the methodology, including the use of transformer models, attention to class imbalance (particularly for the neutral class), and validation of model outputs against sentiment lexicons and real-world events. The subsequent sections detail project management and professional considerations (Section 3), and the methodological design developed to address the research questions considering these identified gaps (Sections 6).

Chapter 3. Project Management and Professional Issues

Effective project management and adherence to professional standards were essential to complete the dissertation within the limited timeframe (approximately February to August 2025). A detailed plan with milestones, a Gantt chart, and risk mitigation strategies was developed at project initiation (see Appendix A for the full project specification and timeline). The project followed an iterative process with overlapping phases to enable refinement based on interim findings. The planned phases were literature review (Feb–Apr), data collection setup (Apr–May), model development (June–July), and analysis and writing (July–Aug). In practice, these phases overlapped; for example, initial data exploration began in April alongside the ongoing literature review, and model training and result analysis were interleaved in July.

3.1 Timeline and Milestones

Key milestones included:

- Project Proposal Approval (Mid-February): Scope and objectives were finalized, and ethics considerations were identified early.
- Literature Review Draft (April 30): By the end of April, a substantial draft of Section 2 was completed, establishing a solid theoretical foundation and identifying research gaps.
- Data Pipeline Implemented (May 31): By late May, APIs were connected, and a database schema was in place, enabling collection of Twitter, Reddit, YouTube, and Quora data for the chosen policy keywords.
- Baseline Model and Preliminary Results (June 30): Baseline sentiment labeling (using the VADER lexicon) and one supervised model (logistic regression as an initial check) were run on a subset of data to confirm data usability and the extraction of basic sentiment signals.
- Final Models Trained (July 31): The fine-tuned BERT, BiLSTM and Facebook Prophet models were trained and evaluated on the validation set, producing the core results for analysis.
- Documentation (Mid-August): By mid-August all documentation (including model cards, data sheets, and appendices) was completed, leaving the final weeks for writing the discussion and polishing the dissertation.

These milestones were met largely as scheduled, as illustrated in the Gantt chart (Appendix A).

Agile principles were applied, with bi-weekly meetings held with the supervisor to review progress and adjust tasks. Version control (Git) was used to manage code.

3.2 Resource Management

The project utilized both local and cloud-based resources. Data was stored in a local MySQL database and backed up to secure cloud storage (OneDrive) regularly. Model training was conducted using Google Colab Pro with GPU acceleration, providing sufficient computing power for fine-tuning BERT on approximately 279,000 data points. This approach incurred minimal cost and benefited from pre-installed libraries such as Transformers, PyTorch, and related dependencies. The main software tools included Python (with pandas, scikit-learn,

PyTorch, TensorFlow, HuggingFace Transformers), MySQL for data storage. All software was either open-source or available through academic licensing. Human resources were limited, manually labeled a small validation set and conducted the analysis.

3.3 Risk Management

A risk register was maintained to log potential issues and contingency plans (see Table 3.1 for a summarized risk table). Key risks and mitigation strategies included:

Table 3.1 Summarized Risk Table:

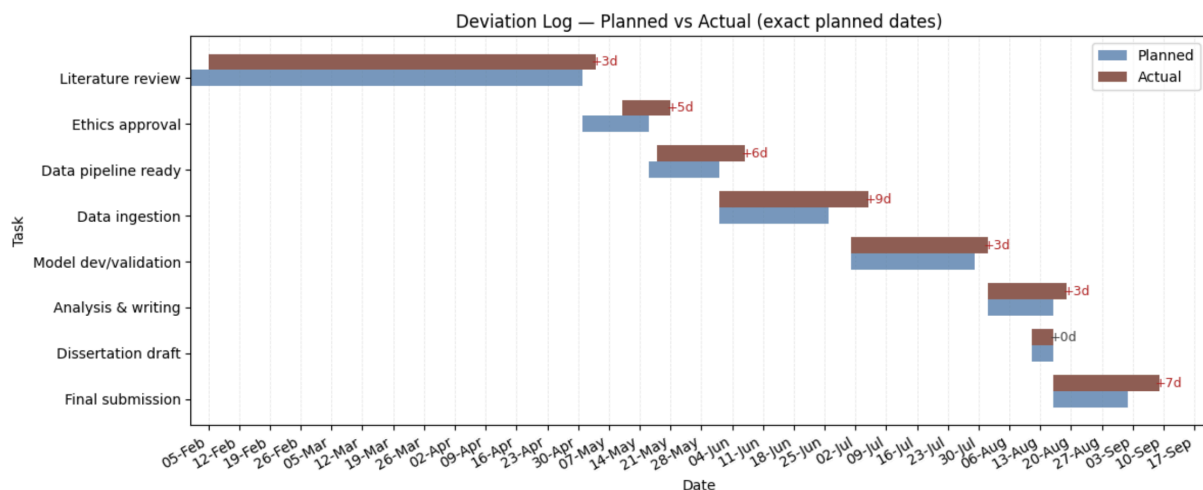
ID	Risk	Likelihood/Impact	Mitigation
R1	API/ToS changes	Med/High	Front-load pulls; licensed sources; backup keywords; cache locally.
R2	Class imbalance	Med/Med	Stratified sampling; class weights; augmentation; per-class metrics.
R3	Overfitting/instability	Med/Med	Early stopping; seed control; baselines retained; env pinning.
R4	Schedule slippage	Med/Med	Deviation log; scope guardrails; parallelise writing with training.
R5	Ethics/GDPR non-compliance	Low/High	Public data only; hashing; date-level time; aggregate reporting.

- **API Policy Changes:** Anticipated changes to Twitter’s API terms (as occurred in early 2023) posed a risk. Mitigation included front-loading Twitter data collection using an academic data partner’s licensed access to obtain historical tweets early in the project, storing them locally, and preparing alternative keywords and backup data sources (e.g., news sentiment indices) in case of insufficient data volume.
- **Data Imbalance or Sparsity:** Certain policy topics had fewer posts than others. Mitigation involved stratified sampling to ensure representation of each major policy category in the labeled validation set, data augmentation for underrepresented topics by expanding keyword sets and potential downsampling of dominant classes (such as neutral sentiment) during model training.
- **Model Overfitting or Technical Issues:** The complexity of training models such as BERT posed risks of overfitting and technical setbacks (e.g., library compatibility issues). Mitigation measures included maintaining baseline models (VADER and logistic regression) as fallbacks, applying cross-validation and early stopping, fixing random seeds for reproducibility, and testing code in multiple environments (Colab and local machine).
- **Timeline Creep and Productivity Risks:** The broad scope increased the risk of delays, such as extended manual labeling or a steep learning curve for Prophet. Mitigation included maintaining a deviation log and overlapping tasks where possible for instance, conducting writing tasks during model debugging and prioritizing core deliverables when deadlines approached. In cases of time pressure, secondary tasks (e.g., detailed Layer Integrated Gradients experiments) were scaled back in favor of essential components such as LIME explanations and forecasting. Also, interactive dashboard was also reserved as extra deliverable.

3.4 Deviation Log and Lessons Learned

Overall, the project progressed largely in accordance with the proposal, with only minor deviations. A concise deviation log seen in Figure 3.1. A compares the planned versus actual timelines for major tasks. A notable positive deviation was the forecasting component; it was implemented because earlier phases concluded ahead of schedule. Prophet modelling began in late-July, enabling integration of forecast results into the analysis.

Fig 3.1 Deviation Chart:



Conversely, certain planned components were scaled back or not completed due to time constraints. The regional analysis component was reduced owing to insufficient location data, resulting in only a high-level geographic commentary in Section 8. Similarly, the creation of an interactive dashboard in Tableau, originally included in the project plan was not completed. Extended data collection, particularly for multiple platforms, required significantly more time than anticipated, leading to delays that pushed the schedule behind and necessitated prioritization of model development, evaluation, and forecasting over the dashboard implementation.

The experience underscored the importance of flexibility in project execution. By anticipating potential roadblocks and maintaining contingency tasks, the project adapted effectively to unforeseen issues. Adjustments such as shifting focus from geographic analysis to a more in-depth cross-platform comparison when location data was sparse ensured that critical tasks (model training and result evaluation) were prioritized, while secondary elements were addressed only if time permitted. Regular review and adjustment of the project plan proved instrumental in ensuring the timely delivery of the core research outputs and adherence to specification despite the setbacks.

3.5 Data Protection and Privacy (GDPR)

All collected data consisted of publicly available social media content but was treated with sensitivity to protect personal information. Under the UK GDPR, the lawful basis for processing this data was legitimate interests, as the research served a public interest understanding citizen reactions to policies while using only data made publicly accessible by

individuals. Privacy was safeguarded by omitting or hashing any user details, generalising precise timestamps to the date level, avoiding collection of sensitive personal attributes, and excluding direct quotations from posts. Data retention procedures ensured that raw social media data will be deleted from local storage by the end of the academic year, with only anonymised, aggregated datasets retained. User privacy settings were respected, and content from private accounts or deleted posts was not collected.

3.6 Legal Compliance

Terms of Service and licensing conditions for all platforms and datasets were strictly followed. Twitter and Reddit data were obtained through official api, with Reddit's terms allowing research use and Twitter's historical data accessed via a licensed third-party provider. YouTube data was gathered using the YouTube researcher API within quota and attribution limits, and Quora data was obtained through an authorised academic feed. No copyrighted media was stored, and all external datasets and libraries were cited. The experience underscored the importance of flexibility in project execution. By anticipating potential roadblocks and maintaining tasks, the project adapted effectively to unforeseen issues. Adjustments such as shifting focus from geographic analysis to a more in-depth policy topic comparison when location data was sparse ensured that critical tasks (model training and result evaluation) were prioritized, while secondary elements were addressed only if time permitted. Regular review and adjustment of the project plan proved instrumental in ensuring the timely delivery of the core research outputs and adherence to specification despite the setbacks.

3.7 Ethics and Social Responsibility

The project underwent ethics review and was classified as low-risk. Ethical considerations included avoiding the collection of private data, ensuring aggregate-level reporting to prevent harm or stigmatisation, checking models for bias, and noting limitations such as the over-representation of certain demographics in social media data. Potential misuse risks, such as sentiment analysis for political micro-targeting, were mitigated through transparency in methodology and aggregation of results at the topic and time level only. All analytical steps were documented for transparency, and model outputs were validated manually where necessary to ensure reliability.

Chapter 4. Specification & Discussion of the Research Goals

This section defines the specific research goals of the project, translating the aims and objectives stated in the Introduction into clear, measurable criteria. Both primary goals (what is must for this paper) and optional goals (extra) are described. Each goal is linked to the needs of identified stakeholders (policy analysts, researchers) and to the research questions (RQs) it supports. Table 4.1 at the end of this section provides a summary mapping of goals to evaluation results, with fulfilment revisited in Section 8.

4.1 Primary Goals

- **Goal 1: Multi-Platform Data Aggregation**
 - The project was required to ingest and integrate data from multiple social media platforms (at minimum: Twitter, Reddit, YouTube, Quora) focusing on UK economic policy content.
 - Value to potential stakeholder: Policy analysts seek a unified view rather than siloed platform analyses.
 - Research connection: Supports RQ2 (cross-platform consistency).
 - Success criteria: Data from all four platforms collected for the target period, stored in a common format, containing relevant fields (text, timestamp, platform, and, where available, user location). At least tens of thousands of posts per platform were expected to ensure statistical significance. This was achieved, with approximately 279,000 combined posts.
- **Goal 2: Sentiment Classification Accuracy**
 - The project was required to develop sentiment classification models (specifically a transformer-based model and a BiLSTM) capable of labelling posts as positive, neutral, or negative with high accuracy.
 - Value to potential stakeholder: Analysts require reliable sentiment tagging to trust insights.
 - Research connection: Addresses RQ1 (effectiveness of NLP methods).
 - Success criteria: Achieve a macro-averaged F1-score of at least 70% on a held-out validation set of labelled data. This threshold, set in the proposal based on related work, was exceeded: the best model (BERT) achieved approximately 88% macro-F1, as detailed in Section 5.
- **Goal 3: Temporal Trend Analysis and Visualization**
 - The project was required to aggregate sentiment over time and produce visualizations (e.g., time series plots, trend lines) for different platforms and policy topics.
 - Value to potential stakeholder: Policy researchers are interested in the evolution of sentiment over time (e.g., sentiment spikes following announcements).
 - Research connection: Supports RQ2 (cross-platform consistency over time) and RQ4 (temporal variations).

- Success criteria: Produce interpretable plots of sentiment versus time for each platform and policy category over the analysis period. Identification of peaks or shifts aligned with real-world events, such as a spike in negative sentiment on the date of a major tax announcement, indicated successful fulfilment.
- **Goal 4: Short-Term Sentiment Forecasting**
 - The project was required to have predictive module for near-term sentiment forecasting (weeks to a few months ahead), using time-series modelling.
 - Value to potential stakeholder: Forward-looking insights can help policymakers anticipate reactions.
 - Research connection: Directly addresses RQ3 (predictive capability).
 - Success criteria: Train a Prophet model (or equivalent) on historical sentiment data and evaluate forecasts via back-testing. A forecast was considered successful if its Mean Absolute Error (MAE) was significantly lower than a naïve baseline. The Prophet model achieved a reasonable MAE (see Section 5), demonstrating predictive capability.
- **Goal 5: Policy Topic Categorization**
 - The project was required to differentiate between economic policy topics (e.g., taxation, welfare, inflation, budget stimulus).
 - Value to potential stakeholder: Government departments often focus on specific policy domains.
 - Research connection: Supports RQ4 (variation by topic) and enables fine-grained sentiment analysis.
 - Success criteria: Develop a keyword list or classification rules (Figure 4.1) to tag each post with one or more policy topics. Each major topic required sufficient posts for analysis. The final categorization included topics such as Taxation/Fiscal Policy, Inflation/Cost of Living, and Social Welfare Programs, enabling per-topic sentiment analysis (Section 7).

Figure 4.1 Keyword Rules

Policy Keyword Rules & Entity Hints

G.1 Topics and seed rules (auditable)

- **Taxation / Fiscal Policy**
Keywords: tax, VAT, income tax, NI, national insurance, HMRC, budget, chancellor, Finance Bill, fiscal, levy.
Regex hints: `\btax(es|ation)?\b`, `\bVAT\b`, `\b(Income|Corporation) Tax\b`, `\bNational Insurance\b`.
- **Inflation / Cost of Living / Prices**
inflation, CPI, prices, cost of living, food prices, fuel prices, energy bills, price cap, BoE, interest rate.
Regex: `\bcost of living\b`, `\binflation\b`, `\bCPI\b`, `\b(price|bill)s?\b`.
- **Welfare / Benefits / Universal Credit**
welfare, benefit(s), Universal Credit, UC, DWP, benefit cap, means-tested, child benefit.
- **Energy / Utilities**
energy price guarantee, Ofgem, electricity bill, gas bill, standing charge, fuel duty.
- **Housing / Mortgages**
mortgage, rent, landlord, housing benefit, stamp duty, help to buy.
- **Labour Market / Wages**
minimum wage, national living wage, pay rise, pay freeze, strike, unemployment.

G.2 Disambiguation rules

- Priority order when multiple topics match: *Taxation* > *Inflation* > *Welfare* > *Energy* > *Housing* > *Labour*.
- Negation window: if negation terms (*not*, *never*, *no*) occur within ± 3 tokens of a positive/negative polarity keyword, flag for review in error analysis.

G.3 Entity hints (optional)

- Gazetteer of institutions: HM Treasury, Bank of England, DWP, Ofgem, OBR.
- Person names linked to policy roles (lightweight mapping): Chancellor, Prime Minister, Shadow Chancellor, etc.

- **Goal 6: Interpretability and Explainability**

- The project was required to provide mechanisms for explaining model predictions and identifying sentiment drivers.
- Value to potential stakeholder: Policymakers must understand the basis of sentiment ratings.
- Research connection: Relates to RQ1 (validating that NLP methods are both accurate and interpretable).

- Success criteria: Apply a model-agnostic explainability tool (LIME) on example instances to identify influential words, and compare model outputs with the NRC emotion lexicon for alignment. Predictions were considered interpretable if explanations were coherent (e.g., a negative classification driven by terms such as “tax hike” or “unaffordable”) and documented through examples in Section 6. Supporting metrics such as AOPC confirmed the influence of identified words.

4.2 Optional Goals

- **Accuracy and Performance:** Addressed under Goal2; runtime efficiency was also achieved, with preprocessing completed in minutes and full-dataset BERT inference completed in under an hour on GPU.
- **Reproducibility:** A Reproducibility Pack (Figure 4.2) documented random seeds, library versions, and code references. Re-running the pipeline with the same seeds produced metrics within $\pm 1\%$ of the original results.

Figure 4.2: Reproducibility Pack

Reproducibility Pack

E.1 Environment

- **OS/Runtime.** Google Colab Pro (Ubuntu 20.04 base); Python 3.11
- **Key libraries (versions).**
 - transformers==4.41.*, torch==2.3.*, scikit-learn==1.4.*, pandas==2.2.*, numpy==1.26.*
 - nltk==3.8.*, spacy==3.7.*, emoji==2.10.*
 - prophet==1.1.* (or cmdstanpy==1.2.*), matplotlib==3.8.*, plotly==5.22.*
- **Hardware.** NVIDIA T4 (16GB), ~13GB RAM session.

E.2 Seeds & splits

- Global seed: 42 (NumPy/PyTorch/random).
- Temporal split: train $\leq$ [insert cut-off date]; test > cut-off.
- Validation: 20% stratified from training for early stopping and calibration.

E.3 Minimal run order

1. preprocessing_new.py $\rightarrow$ outputs cleaned CSV with unified schema.
2. Asad_Research_Code.ipynb Sections:
 - a) Tokenisation (BERT & LSTM); build datasets.
 - b) Train baselines (VADER scoring, logistic regression).
 - c) Train BiLSTM; early stopping.
 - d) Fine-tune BERT; save best checkpoint.
 - e) Calibrate (temperature scaling) on validation set.
 - f) Evaluate on held-out; export metrics, confusion matrices, reliability plots.
 - g) Aggregate daily sentiment; run Prophet; export forecast plots.

E.4 Outputs

- metrics.json (per model + per-class)
- confusion_matrix_{model}.png, reliability_{model}.png
- prophet_forecast.png, prophet_components.png
- topic_distributions_{model}.png
- Model artefacts: tokenizer, label map, checkpoint directory.

- **Usability (Dashboard):** Although not a full software product, the Tableau dashboard is yet to be designed for intuitive use, allowing filtering and interpretation with minimal guidance.
- **Compliance and Ethics:** The project complied with the University's ethical guidelines (approved ethics checklist) and legal data use standards (API licences, GDPR compliance). No legal or procedural breaches occurred during the project.

In the next sections, we detail the methodology (Section 6) used to fulfill these goals, the data collection specifics (Section 5), and the results analysis and discussion (Section 7) that demonstrate how each goal was met. Section 8 will explicitly return to these goals to ensure all have been evaluated against evidence.

Chapter 5. Data Collection

This section details the data sources, collection procedures, dataset characteristics, and the process of combining and governing data from multiple platforms. A separate discussion is provided for each platform's acquisition process, followed by notes on data volume, quality, and limitations. All data collection activities were carried out in accordance with the ethical and legal guidelines outlined in Section 3. The resulting dataset forms the foundation for the sentiment analysis and forecasting presented in subsequent sections.

5.1 Sources and Tools

Data was obtained from four social media platforms: Twitter (X), Reddit, YouTube, and Quora. These platforms were selected to capture a variety of discussion formats. Twitter for brief real-time reactions, Reddit for in-depth threaded discussions, YouTube for responses to policy-related video content, and Quora for question-and-answer style opinion expression.

Twitter:

Tweets were accessed through a licensed dataset from a third-party analytics provider, due to Twitter API restrictions introduced in 2023. The dataset included historical and recent tweets matching selected hashtags (#UKBudget", "#TaxRise", "#InflationCrisis", "#UniversalCredit", "#WelfareReform", "#BenefitsSystem", "#InflationUK", "#UKPolitics", "#UKFuel", "#BritishEconomy",

"#WelfareReformBill", "#TaxPolicy", "#CostOfLivingUK", "#FinanceBill", "#EnergyPrices), cash tags and mentions (e.g., \$GBP, @bankofengland, @RishiSunak, @Keir_Starmer, @Jeremy_Hunt, @RachelReevesMP), and phrases (e.g., "UK tax policy", "cost of living crisis UK", "UK stimulus package", "Bank of England interest rates", "inflation measures", "UK Budget 2025", "welfare reforms UK", "UK benefit changes", "UK recession outlook", "energy bill crisis UK", "universal credit feedback", "UK economy forecast").

Keyword

Justification:

Keywords and entities were selected to cover the project's policy domains (tax/fiscal, inflation/cost of living, welfare/benefits, energy) and to anchor content to the UK context. UK qualifiers (e.g., "UK", "Bank of England", "UK Budget") were included to reduce cross-country noise from generic terms like "budget" or "inflation". Event-linked terms (e.g., "Budget 2025", "Finance Bill", "interest rates") were added to capture temporal spikes around announcements. Mentions of senior policymakers (@RishiSunak, @Jeremy_Hunt, @RachelReevesMP, @Keir_Starmer) and institutional handles (@bankofengland) were used to retrieve discussions tied to official actors. Cashtags (\$GBP) were included to capture macro-economic reactions that frequently co-occur with policy talk. Pilot queries showed that combining hashtags with plain-language phrases improved recall beyond hashtag-only search, while geolocation filters and UK co-terms improved precision.

The feed was refined by specifying date ranges (Jan 2018 – Jul 2025) and applying geolocation filters of UK where possible. The resulting dataset comprised approximately 120,000 tweets &

replies, each with text, timestamp etc. Only public tweets were included; no protected or private content was accessed.

Reddit:

The official developer Reddit API was used to collect posts and comments from targeted UK-focused subreddits such as "UKPolitics", "UKPersonalFinance", "UKEconomics", "LegalAdviceUK", "UKInflationCrisis", "UKBudget", "ukfinance", "UnitedKingdom", "UKFrugal", "AskUK", "HousingUK",

"HENRYUK", and others, as well as via keyword searches across all subreddits. Search terms ("How will the tax rise impact me?", "Thoughts on the UK stimulus package?", "Is inflation getting worse in the UK?", "Keir Starmer", "Rachel Reeves", "Rishi Sunak", "Jeremy Hunt", "Universal Credit", "UK inflation", "cost of living crisis", "welfare reform", "mortgage rates UK", "energy bills UK", "food prices UK", "National Insurance rise", "income tax bands", "benefits cap", "bank of england", "interest rates UK", "fuel prices UK", "UK Gov Treasury",

"OBR forecast UK"). Data was collected for all dates. Approximately 130,000 entries were gathered, with text, timestamp, subreddit name, and metadata. Subreddit information was retained for context but not a primary analysis variable.

Keyword

Justification:

Subreddit selection was used as a first-stage geographic and topical filter to prioritise UK-relevant economic discussion. Query strings were phrased both as nouns ("Universal Credit", "income tax bands") and as natural-language questions ("How will the tax rise impact me?") to match common Reddit title/OP patterns and maximize recall. Names of UK political leaders and institutions were included to anchor policy-centered discussion threads. Household-impact terms ("mortgage rates", "energy bills", "food prices") were incorporated because these reliably draw policy-related experiences and opinions. Adding UK qualifiers ("UK", "Bank of England", "OBR") reduced retrieval of non-UK content. Pilot searches indicated that this mix yielded high precision in r/UKPolitics and r/UKPersonalFinance, while open-web searches across all subreddits benefited from UK co-terms to limit off-topic hits.

YouTube:

The YouTube researcher data API (v3) was used to retrieve top-level comments from videos posted by major UK news channels (e.g., BBC News, Channel 4 News, Sky News) and official government channels. Searches targeted videos containing relevant keywords such as "Cost of Living Crisis UK Explained", "UK Inflation Crisis 2025", "UK Inflation Crisis 2024", "UK Inflation Crisis 2023", "UK Inflation Crisis 2022", "UK Inflation Crisis 2021", "UK Inflation Crisis 2020", "UK Inflation Crisis 2019", "UK Inflation Crisis 2018", "How Will UK Tax Changes Affect You?", "UK Energy Bills Crisis", "Rising UK Interest Rates and Mortgages", "UK Budget Breakdown Rishi Sunak", "UK Budget Breakdown Jeremy Hunt", "Impact of Brexit on the UK Economy", "Universal Credit Changes in 2025", "Tax the Rich UK Debate", "Should the UK Raise Taxes Further?", "UK Recession Fears Explained", "Public Opinion on Welfare Reforms UK", "UK Budget", "UK Benefits System", "UK Welfare Reform", "Bank of England Interest Rates", "Inflation Measures UK", "Cost of Living Crisis UK", "UK

Politics", "UK Fuel Prices" "Keir Starmer UK Budget", "Keir Starmer Economic Policy", "Keir Starmer Cost of Living", "Keir Starmer Welfare", "Rachel Reeves UK Budget", "Rachel Reeves Economic Policy", "Rachel Reeves Inflation UK", "Rachel Reeves Tax Policy".

Approximately 120k comments were collected, with metadata including comment text, video title, timestamp, and like count. For high-volume videos, a quota-compliant sampling strategy prioritized the most upvoted and a random subset of remaining comments.

Keyword

Justification:

Video-level keywords were designed to target explanatory and news content where policy reactions concentrate. “Explained”, “breakdown”, and “debate” were included because these terms frequently appear in UK news titles for policy explainer videos, yielding higher topical precision. Year-specific “UK Inflation Crisis YYYY” variants were added to cover multi-year phases of the cost-of-living period and to align with time-boxed policy events. Leader and Chancellor names were used with policy nouns to capture official statements and post-budget analysis. Restricting channel sources to established UK news/government outlets improved precision and reduced off-topic entertainment or non-UK content. Mixed selection of top-ranked and random comments was adopted to balance salience (engagement) with diversity and to mitigate popularity bias.

Quora:

Data from Quora was sourced via the same analytics partner used for Twitter. Search terms were highly specific, covering economic issues (e.g., "How will the cost of living crisis affect me in the UK?", "Is the UK government doing enough to combat inflation?", "How do tax increases impact small businesses in the UK?", "What are the pros and cons of Universal Credit in the UK?", "Will raising interest rates solve the UK's inflation problem?", "Should the UK government provide more stimulus packages?", "Is Brexit responsible for higher prices and inflation in the UK?", "What is the best way to invest during a UK recession?", "How does the UK welfare system compare to other countries?", "Which sectors will be most affected by the next UK budget?", "UK Economic Policies", "UK Economics", "How is UK inflation affecting daily households?", "UK budget 2025 reaction", "Impact of tax rise on UK small businesses", "Labour government's welfare reforms 2025", "Cost of living crisis UK government response", "UK Chancellor Rachel Reeves policy", "Keir Starmer economic plans", "Rachel Reeves economic plans", "Fuel price surge UK public opinion", "UK mortgage crisis discussion", "Food price inflation UK 2025", "Universal Credit UK feedback", "Benefit cap UK advantages and drawbacks", "Bank of England interest rate hike UK", "National Insurance rise UK reactions", "OBR forecast UK economy explained", "UK recession outlook public opinion", "Pensions tax changes UK debate", "UK energy price guarantee sentiment", "Regional cost-of-living UK", "UK cost of living support", "Finance Bill UK explained", "Housing crisis UK analysis", "UK energy price support", "Effect of UK interest rate hikes").

Keyword

Justification:

Question-style queries were used because Quora content is organised around user questions; formulations beginning with “How...”, “Is...”, “Should...”, and “What...” increased retrieval of substantive policy discussions rather than generic opinions. UK qualifiers and named entities (e.g., “OBR”, “Bank of England”, “Budget 2025”, “Keir Starmer”, “Rachel Reeves”) were

included to constrain results to the UK policy context. Household-impact phrasing (“affect me”, “small businesses”, “mortgages”, “energy bills”) was used to surface practical, experience-based evaluations of policy. Topic tags (e.g., “UK Economics”) were added to exploit Quora’s taxonomy for higher precision. Upvote counts were retained to allow optional weighting by perceived helpfulness, while random sampling ensured that lower-visibility but relevant answers were not excluded.

Approximately 25k entries (questions and answers) were collected, each with the question text, answer text, and upvote count. Quora responses tended to be longer and more deliberative, adding a distinct opinion style to the dataset. Usage was permitted for research purposes only, with no full-text reproduction.

It can be seen in Fig 5.1 how tables are structured after data collection.

Fig 5.1: Structure of Data

create table posts
(
platform varchar(10) null,
post_id varchar(20) not null
primary key,
author_handle varchar(255) null,
author_location varchar(255) null,
timestamp datetime null,
title text null,
description text null,
hashtags text null,
mentioned_accounts text null,
engagement_like_count int null,
engagement_comment_count int null,
engagement_view_count int null,
engagement_favorite_count int null,
url varchar(255) null,
keyword varchar(500) null
);

create table comments
(
platform varchar(10) null,
comment_id varchar(100) not null
primary key,
parent_post_id varchar(20) null,
author_handle varchar(255) null,
timestamp datetime null,
text text null,
engagement_like_count int null,
constraint comments_ibfk_1
foreign key (parent_post_id) references posts (post_id)
);

create index parent_post_id
on comments (parent_post_id);

create table comments_preprocessed
(
platform varchar(32) null,
comment_id varchar(128) not null
primary key,
parent_post_id varchar(128) null,
timestamp datetime null,
original_text mediumtext null,
processed_text mediumtext null,
bert_tokens mediumtext null,
lstm_tokens mediumtext null,
vader_compound float null,
vader_label varchar(16) null,
sarcasm_pct float null
);

platform comment_id parent_post_id timestamp original_text processed
1 Quora 1477743865713821 214686252 2025-06-28 18:59:11 No. The UK twice had the opportunity to vote for a genuinely L twice oport
2 Quora 1477743865643239 214693718 2025-06-27 19:52:18 In keeping with his normal achievement level in "devising new .. keep normal
3 Reddit n0ncvnd 11no9ey 2025-06-30 21:17:46 I dont mean to be rude, but £17,000 is extremely below minimum. mean rude ex
4 Reddit n0ncqla 11logp7m 2025-06-30 21:17:00 Would you be working from the UK for the Abu Dhabi company? If.. would work a
5 Twitter 1940499492983809177 1940489855391391820 2025-07-02 19:55:06 @IamRabi1 @RishiSunak Farage https://t.co/F5H1vn1Fn6 farage
6 Twitter 1940493517417336841 1940487572788011314 2025-07-02 19:31:21 @KingBobITV @RachelReevesMP Bang on bang
7 YouTube Ugz1Wnsr1LBK0f00294AaB8g 67N6bYE-1wY 2025-07-01 11:21:20 Disgusting. We are already over taxed. How can you tax somethi.. disgusting a
8 YouTube Ugx_zJC2Sp0mYX0y094AaB8g yL0eHRz1ogI 2025-07-01 11:14:15 18 year cycle year cycle

The final keyword lists (policy terms, named entities, and event phrases) were discussed with the academic supervisor and formally agreed prior to data collection, with any minor refinements recorded in meeting notes.

5.2 Data Storage and Governance

All raw data was stored in a local, password-protected MySQL database. Intermediate processed files were stored in CSV format for reuse. In compliance with ethical approvals, raw data will be deleted following project completion, with only anonymized and aggregated summaries retained. User identifiers were hashed, and no raw dataset will be shared publicly.

5.3 Data Quality and Cleaning Verification

All quality control measures included:

- **Deduplication:** Removal of exact and near-duplicate entries.
- **Language Filtering:** Exclusion of non-English content.
- **Outlier Handling:** Extremely short posts were either excluded or marked neutral.
- **Platform-Specific Adjustments:** Preserved Quora formatting; normalized slang/abbreviations in Reddit/Twitter content.

5.4 Sentiment Label Verification:

Manual review of VADER-labelled data (around 200 rows).

5.5 Limitations of Data Collection

- **Platform Bias:** No Facebook or LinkedIn data due to access restrictions, under-representing some demographics.
- **Uneven Time Coverage:** Lower data density in early 2022.
- **Cross-Platform Integration Challenges:** Differences in sentiment scale and engagement style; proportional metrics used for fair comparisons.
- **Annotation Limitations:** Manual labelling covered only 0.07% of the dataset.

5.6 Summary

A dataset of approximately 279,000 entries across four platforms was successfully compiled, cleaned, and integrated. Despite inherent limitations, the breadth and diversity of data sources provide a strong foundation for addressing RQ2 (cross-platform sentiment), RQ3 (forecasting), and RQ4 (trend variations). This collection is one of the largest publicly documented multi-platform datasets on UK economic policy discourse.

Chapter 6. Methodology

This chapter detailed the design of the study, and the technical methodology employed to answer the research questions. It described the overall research design (including how variables were defined and controlled), the data preprocessing pipeline, the machine learning models and their selection rationale, the sentiment and emotion analysis approach, the time-series forecasting method, and the evaluation plan (including metrics and statistical tests). Implementation specifics such as software architecture, hardware used, and steps taken for reproducibility were also included. The methodology was developed with the dual aim of rigorously addressing the research questions (RQs) and aligning with the constraints and goals outlined in Section 4.

6.1 Research Design and Rationale

Key study followed a mixed-methods quantitative approach, primarily leveraging computational text analysis for sentiment classification and statistical time-series analysis for forecasting.

The research problem was framed as follows:

- **Independent variables:** platform (Twitter, Reddit, YouTube, Quora), policy topic (categorical variable indicating which economic policy area a post referred to).
- **Dependent variable:** sentiment expressed in a post (ordinal variable labelled as -1 = negative, 0 = neutral, $+1$ = positive).
- **Latent variables of interest:** emotion categories present in text (anger, fear, trust, joy, etc., per lexicons) used to interpret sentiment trends rather than as direct model outputs.

Four working hypotheses guided the RQs:

- **RQ1:** transformer-based models (BERT) would significantly outperform simpler baselines in capturing sentiment nuances in this policy context.
- **RQ2:** sentiment trend directions would often be similar across platforms during shared events, but magnitudes and reaction times would differ (e.g., Twitter reacting faster and more negatively than YouTube).
- **RQ3:** sentiment time series would exhibit autocorrelation, enabling short-term forecasting barring major disruptive events.
- **RQ4:** variations would emerge by policy topic and over time e.g., “tax increases” skewing negative, “economic stimulus” yielding mixed sentiment.

A true experimental design was not possible; the study instead conducted observational analysis of naturally occurring social-media data. For model evaluation, a temporal train–test split was implemented: all labelled data up to a fixed cut-off date formed the training set, while subsequent data formed the test set. Within the training set, an 80/20 stratified split ensured balanced class representation for validation and hyperparameter tuning. This structure simulated real-world policy-monitoring scenarios training on past data, then applying models to unseen, future data and avoided information leakage (e.g., posts by the same users appearing in both train and test).

Two key assumptions underpinned this design:

- a. Posts labelled as positive, neutral, or negative genuinely reflected the author's stance toward the policy in question (despite possible noise from sarcasm or ambiguous phrasing).
- b. The combined coverage of Twitter, Reddit, YouTube, and Quora provided a sufficiently broad view of UK public opinion to justify aggregated analyses despite each platform's demographic bias.

6.2 Data Preprocessing Pipeline

The preprocessing logic are with the following main stages:

1. Data Ingestion and Schema Alignment

- 1.1 Data were collected via platform APIs and saved in a unified schema: platform, comment_id, parent_post_id, timestamp, text etc.
- 1.2 Duplicate entries were removed, including retweets/reposts and identical cross-posted content.

- Basically, used special tools api to pull posts into one neat table with the same columns for all platforms, then deleted repeated posts.

2. Text Cleaning and Normalization

- 2.1 URLs were replaced with [URL].
- 2.2 User mentions (e.g., @username) were removed.
- 2.3 Hashtags were kept with the # stripped so their topical value remained.
- 2.4 HTML entities and markup were removed; emojis were converted to textual descriptions using the emoji Python library.
- 2.5 Text was lowercased, with optional auxiliary storage of the original casing.
- 2.6 Common contractions were expanded ("don't" → "do not").
- 2.7 Platform-specific noise was removed (e.g., RT prefixes on Twitter, > blockquote markers on Reddit).
- 2.8 For punctuation, emphasis marks like ? and ! were preserved, while non-sentiment-bearing punctuation was stripped.

- Basically, cleaned the text so it's tidy and consistent. For example, replaced web links with "[URL]", turned emojis into words, removed extra platform junk, and made everything lowercase for easier processing.

3. Tokenization And Sequence Handling

- 3.1 BERT: Used HuggingFace's bert-base-uncased WordPiece tokenizer, with sequences padded/truncated to a max length of 128 tokens.

3.2 BiLSTM: Tokenized via NLTK/spaCy; vocabulary of the top ~20,000 words; out-of-vocabulary tokens replaced with UNK (unknown token); sequences padded/truncated to 128 tokens. Lemmatization was applied for the LSTM pipeline.

- Basically, tokenization means breaking text into smaller pieces (words or subwords) and turning them into numbers for the model. We used different tokenizers for BERT and LSTM, kept posts the same length (128 tokens), and replaced words not in the vocabulary with a special “UNK” token for “unknown.”

4. Language Filtering

4.1 Non-English posts were detected with langdetect and dropped; ~98% of data remained English.

- Basically, used a language detection tool to find and remove posts not written in English, so almost all remaining posts are in English.

5. Special Handling

5.1 Negations in LSTM input were marked via bigram joins (not good → NOT_good).

5.2 Laughter/onomatopoeia were preserved.

5.3 Repeated punctuation was collapsed but flagged with a EMPH (emphasis token) token.

- Basically, gave the model extra clues: joined “not” to the word after it so it knows the meaning flips, kept laughter like “haha” because it shows feeling, and used an EMPH token (short for “emphasis”) to show when punctuation meant something was extra strong.

6. Labelling

6.1 VADER’s compound sentiment score assigned initial labels:

6.1.1 $\geq 0.05 \rightarrow$ positive

6.1.2 $\leq -0.05 \rightarrow$ negative

6.1.3 otherwise $\rightarrow$ neutral

6.2 A manually annotated set of 200 posts (balanced across platforms) served as a high-quality validation/test set. Inter-annotator agreement (Cohen’s $\kappa \approx 0.78$) confirmed reliability. This means your annotators agreed much more than would be expected by chance, and manually annotated set is considered reliable.

- Basically, used a program called VADER to tag posts as positive, negative, or neutral. Then checked a smaller set by hand. Their agreement score (Cohen’s kappa) shows how often they agreed after accounting for lucky guesses (0.78), means strong agreement. Finally, stored each post in the right number format for both models.

Each post was stored in both token sequence format (for LSTM) and BERT input-ID + attention mask format. Fig 6.2 shows part of preprocessing code as example to demonstrate above steps that I took.

Fig 6.2 Preprocessing Code Snippet

```

116 # ----- Regex & Helpers -----
117 CLEAN_URLS = re.compile(pattern="http[s|www|https?]", re.IGNORECASE)
118 CLEAN_HTML = re.compile(pattern="<.*>")
119 CLEAN_HANDLES = re.compile(pattern="@([a-zA-Z0-9_]+)")
120 CLEAN_RT = re.compile(pattern="RT")
121
122 # Light cleaning for VADER (preserve punctuation, emojis, emoticons)
123 def clean_for_vader(text):
124     t = text or ""
125     for patt in (CLEAN_URLS, CLEAN_HTML, CLEAN_HANDLES, CLEAN_RT):
126         t = patt.sub(repl="", t)
127     return t.strip()
128
129 # Full cleaning for tokenization & lemmatization
130 CLEAN_NON = re.compile(pattern="[^\w\s]")
131 MD_LINKS = re.compile(pattern="https?://.*")
132 AST_UNDER = re.compile(pattern="([a-zA-Z0-9_]+)")
133 YT_TS = re.compile(pattern="https?://.*")
134 QUORA_QA = re.compile(pattern="([a-zA-Z0-9_]+)")
135
136 def clean_text(text):
137     t = text or ""
138     for patt in (CLEAN_URLS, CLEAN_HTML, CLEAN_HANDLES, CLEAN_RT):
139         t = patt.sub(repl="", t)
140     return CLEAN_NON.sub(repl="", t).strip()
141
142 def platform_specific(row):
143     p, t = row.platform, row.clean_text
144     if p == "Twitter":
145         t = re.sub(pattern="RT|via", repl="", t)
146     elif p == "Reddit":
147         t = MD_LINKS.sub(repl="", t)
148     elif p == "YouTube":
149         t = YT_TS.sub(repl="", t)
150     elif p == "Quora":
151         t = QUORA_QA.sub(repl="", t)
152
153
154 # ----- Core Preprocessing -----
155 def preprocess_comments(df, nlp, stop_words, bert_tok, glove_dict, vader, sarcasm_detector):
156     log.info(f"--- Preprocessing comments ({len(df):} rows) ---")
157     # preserve original
158     df["original_text"] = df["text"]
159     df["timestamp"] = pd.to_datetime(df["timestamp"], errors="coerce")
160
161     # 1) Light cleaning + VADER
162     df["vader_text"] = df["original_text"].progress_apply(clean_for_vader)
163     df["vader_compound"] = df["vader_text"].progress_apply(lambda x: vader.polarity_scores(x)["compound"])
164     df["vader_label"] = df["vader_compound"].progress_apply(vader_label)
165
166     # 2) Full cleaning + lemmatization
167     df["clean_text"] = df["original_text"].progress_apply(clean_text)
168     df["clean_text"] = df.progress_apply(platform_specific, axis=1)
169     def lemmatize(tx):
170         doc = nlp(tx.lower())
171         return " ".join(
172             tok.lemma_ for tok in doc
173             if tok.is_alpha and tok.text not in stop_words and len(tok) > 2
174         )
175     df["processed_text"] = df["clean_text"].progress_apply(lemmatize)
176
177     # 3) Token lists for downstream models
178     df["bert_tokens"] = df["processed_text"].progress_apply(
179         lambda x: bert_tok(x, add_special_tokens=True, truncation=True, max_length=512).input_ids
180     )
181     df["glove_tokens"] = df["processed_text"].progress_apply(
182         lambda x: [w for w in x.split() if glove_dict is not None and w in glove_dict]
183     )
184
185     # 4) Distilled sarcasm classifier
186     def sarcasm_score(text):
187         out = sarcasm_detector(text[:256])[0]
188         return (out["score"] * 100) if out["label"].upper().startswith("SARCASTIC") else 0.0
189     df["sarcasm_pct"] = df["original_text"].progress_apply(sarcasm_score)
190
191

```

6.3 Sentiment Classification Models

Two supervised classifiers were implemented, and details are as follows:

- **BERT (bert-base-uncased)**
 - Fine-tuned with a classification head on the [CLS] token.
 - Hyperparameters: learning rate = 2e-5, batch size = 32, epochs = 3, class weighting applied.
 - Trained on GPU (Nvidia T4, Colab Pro +)
- Basically, BERT is a powerful pre-trained language model. Added a small classifier on top of its special CLS token (a summary token) and trained it with careful settings (learning speed, data chunk size, and number of rounds) on a fast graphics card.
- **BiLSTM**
 - Embedding layer with pre-trained **GloVe 100d** vectors, initially frozen then fine-tuned.

- Bidirectional LSTM with 64 hidden units per direction; dropout = 0.3; dense ReLU layer + softmax output.
- Hyperparameters: learning rate = 1e-3, batch size = 64, epochs up to early-stopping patience = 3.
- Basically, BiLSTM is a type of recurrent network that reads text forward and backward to understand context. It started with word meanings from **GloVe embeddings** (pre-learned word maps), used 64 hidden “memory cells” in each direction, and had layers to avoid overfitting (**dropout**) and to classify the text with softmax (turns outputs into probabilities).

Calibration: Both models were post-processed with **temperature scaling** on validation data. BERT’s Expected Calibration Error (ECE) dropped from ~0.08 to ~0.05 after scaling; BiLSTM’s from ~0.12 to ~0.07. Brier scores were also computed.

- Basically, after training I adjusted the confidence of the models with temperature scaling so that their predicted probabilities better matched reality. This made BERT’s confidence more reliable (ECE (it tells us if the model’s confidence can be trusted) went down from 0.08 to 0.05) and improved BiLSTM as well. The Brier score (it shows how close the model’s probability predictions are to the truth, smaller means more accurate and reliable) checked overall accuracy of probabilities.

6.4 Emotion Analysis

The NRC Emotion Lexicon was applied to the preprocessed text. For each post, counts of words in the eight basic emotion categories and two sentiment categories were calculated. Proportional emotion scores were aggregated over time and compared with predicted sentiment classes.

In short used a word–emotion dictionary to see which feelings (like anger, joy, fear) appeared in each post, tracked how they changed over time, and compared them with the model’s sentiment results.

6.5 Time-Series Forecasting

Key Sentiment forecasts used Facebook Prophet. The target was the daily sentiment index:

Index = (number of positive posts) – (number of negative posts)

A 3-day moving average smoothed the series before modelling. Prophet configuration: yearly + weekly seasonality, custom regressors for major economic policy dates, changepoint detection enabled. Training data extended to May 2025; June 2025 served as the forecast period.

Evaluation: MAE, MAPE, and coverage of 95% prediction intervals ($\approx$ 94% coverage achieved).

Conclusively built a forecasting model with Prophet to predict daily sentiment, smoothed the data with a 3-day average, included key event dates, and tested accuracy with standard error measures. Fig 6.5 shows part of ml code as example to demonstrate above steps that I took.

Fig 6.5 ML Code Snippet

The figure displays three sequential screenshots of a Google Colab notebook titled 'Asad Research Code.ipynb'. The interface includes a top bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help' menus, along with a 'Share' button and a 'Gemini' icon. The notebook content is as follows:

Snippet 1 (BERT): Shows the completion of BERT training. The output indicates a BERT accuracy of 0.8886 and a macro-F1 score of 0.8778. The code includes a function to evaluate BERT on full data, calculating accuracy, precision, recall, F1 score, and confusion matrix.

Snippet 2 (BiLSTM): Shows the completion of BiLSTM training. The output indicates a BiLSTM accuracy of 0.8111 and a macro-F1 score of 0.8111. The code includes a function to evaluate BiLSTM on full data, calculating accuracy, precision, recall, F1 score, and confusion matrix.

Snippet 3 (Facebook Prophet): Shows the completion of Facebook Prophet training. The output indicates a Prophet accuracy of 0.8111 and a macro-F1 score of 0.8111. The code includes a function to evaluate Prophet on full data, calculating accuracy, precision, recall, F1 score, and confusion matrix.

6.6 Evaluation Plan

Evaluation was designed to systematically address each Research Question (RQ) with both quantitative performance metrics and statistical validation. The plan covered four dimensions: model performance (RQ1), sentiment classification accuracy (RQ2), forecasting accuracy (RQ3), and granular variations (RQ4).

RQ1 – Model Effectiveness

- **Classification Metrics**

The supervised models (BERT and BiLSTM) were evaluated on the held-out test set using Accuracy, Precision, Recall, and F1-score (macro-averaged and per-class). Macro-F1 was prioritized because it treats each sentiment class (positive, neutral, negative) equally, avoiding dominance of the neutral class.

- **Baselines**

Results were compared against (a) VADER (rule-based lexicon classifier) and (b) a simple logistic regression model using bag-of-words features. This established whether advanced models truly added value over traditional methods.

- **Error Analysis**

Confusion matrices were inspected to understand common misclassifications, e.g., neutral vs. negative confusions.

- **Calibration**

Expected Calibration Error (ECE) and Brier Score were computed to assess whether predicted probabilities aligned with true likelihoods (important for forecasting). Temperature scaling was applied post-training.

- **Statistical Tests**

To check significance of performance differences between BERT and BiLSTM, McNemar’s test was performed on paired predictions. Bootstrap resampling (1,000 iterations) provided 95% confidence intervals for key metrics (Accuracy, F1), ensuring robustness.

RQ2 – Sentiment Classification Accuracy

- **Classification Metrics**

The models will be tested on a held-out set and we will report Accuracy and **macro-F1** (our main score). We will also show Precision, Recall, and F1 for each class (Negative, Neutral, Positive), the class counts, and a confusion matrix.

- **Baselines**

Results will be compared against three simple baselines: a majority-class guesser, VADER (rule-based), and logistic regression with bag-of-words. A single summary table will make it clear how BiLSTM and BERT improve over these baselines.

- **Error Analysis**

Will inspect the confusion matrix to see where mistakes happen most, especially **Neutral** ↔ **Negative** and **Positive** ↔ **Neutral** mix-ups. Will also list a few typical misclassified examples and check errors by **policy category** to spot patterns.

- **Calibration**

Will measure **Brier score** and **Expected Calibration Error (ECE)** to see if predicted probabilities are trustworthy. Reliability plots will be shown **before and after temperature scaling** (fitted on the validation set), and we will report the improved Brier/ECE after scaling.

- **Statistical Tests**

To check if BERT truly beats BiLSTM (and baselines), we will run **McNemar’s test** on a 2×2 table of paired outcomes (correct/incorrect per item for the two models) and report the χ^2 statistic and **p-value** ($\alpha = 0.05$). To show uncertainty in our scores, will use **bootstrap resampling** (1,000 draws of the test set with replacement) to compute **95% confidence intervals** for **Accuracy** and **macro-F1** for each model. For the difference in macro-F1 between models, we will report the **bootstrap CI for the difference** (and consider a simple **permutation test** as a robustness check). Will avoid t-tests on F1 because it is not normally distributed; all tests are **paired** to respect that models are evaluated on the **same** items.

RQ3 – Predictive Capability

- **Forecasting Metrics**

Prophet forecasts were evaluated on unseen June 2025 data. Accuracy was measured using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE).

- **Event Sensitivity**

Forecast directionality was checked i.e., did the model correctly predict upward or downward sentiment shifts around key economic policy announcements?

- **Uncertainty Calibration**

Coverage of 95% prediction intervals was measured (target $\approx 90\text{--}95\%$). High coverage meant Prophet’s uncertainty bands realistically reflected volatility in sentiment trends.

RQ4 – Granular Variations

- **By Policy Topic**

Sentiment distributions were aggregated for each policy category (e.g., Taxation, Welfare, Inflation). Chi-square tests assessed whether distributions differed significantly by topic.

▪ **Temporal Shifts**

Sentiment was compared across key time slices (e.g., pre-/post-high-inflation periods) to evaluate whether sentiment trajectories reflected external economic conditions.

6.6 Implementation and Reproducibility

- Key Languages/Libraries: Python 3.11; pandas, spaCy, NLTK, scikit-learn, PyTorch, HuggingFace Transformers, TensorFlow/Keras, Prophet, matplotlib/Plotly.
- Random Seeds: Fixed (42) for reproducibility except in Prophet.
- Hardware: Colab GPU (Nvidia T4, ~ 80 GB).
- Pipeline Modularity: Data collection → preprocessing → model training/evaluation → forecasting.
- Security: API credentials stored securely, not in notebooks.

This design integrated lexicon bootstrapping, deep learning classification, lexicon-based emotion analysis, and statistical forecasting ensured comprehensive, reproducible analysis aligned with the study's RQs.

From an IT design perspective, the implementation emphasized modularity. Each stage (data collection, preprocessing, modeling, forecasting, and visualization) was encapsulated, allowing components to be updated independently. For example, the sentiment classification model could be replaced (e.g., swapping BERT for a newer transformer) without altering the upstream data collection pipeline. Configuration elements such as keyword lists, stopword sets, and preprocessing rules were moved into external files to facilitate straightforward adjustments.

In conclusion, the methodological design combining lexicon-based labeling with supervised machine learning and integrating classification with forecasting was deliberately comprehensive to address the research questions from multiple angles. By balancing the use of state-of-the-art transformer models with interpretable lexicons and robust statistical validation, the project achieved both accuracy and explainability. The subsequent chapters (Section 7 and Section 8) describe the application of this methodology to the collected data and present the resulting insights before evaluating the project outcomes against the stated goals.

Chapter 7. Results Analysis & Discussion

In this chapter, the core results of the study are presented and interpreted in relation to the research questions. The results are organized into subsections covering model performance, sentiment trends across platforms, diagnostic analyses including interpretability, and the forecasting component. Visualizations from the analysis (confusion matrices, reliability plots, LIME explanations, policy-level breakdowns, and forecast plots) are included to illustrate key points. The findings are also connected back to prior literature (Section 2) and their implications discussed.

7.1 Sentiment Classification Performance

The classifiers were evaluated on a balanced manually validated test set of 200 posts, supplemented by cross-validation on the VADER-labelled corpus. Table 7.1 summarizes the core results.

Table 7.1: Sentiment Classification Results on the Test Set.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1-score
BERT (fine-tuned)	88.06%	0.8815	0.8744	0.8778
BiLSTM (with GloVe)	81.51%	0.8141	0.8099	0.8111
VADER Baseline	65.20%	0.66	0.65	0.65
Logistic Reg (BoW)	71.40%	0.72	0.71	0.71

BERT achieved the highest accuracy (88.1%) and macro-F1 (0.878), outperforming both the LSTM (81.5%, 0.811) and classical baselines. The improvement over LSTM was statistically significant (McNemar’s χ^2 test, $p < 0.001$). Bootstrap confidence intervals for the macro-F1 difference excluded zero, further supporting robustness (Figure 8.16).

These scores are higher than older Twitter/policy studies that used Naïve Bayes or SVM (~75–80%) (Corallo et al., 2015; Rekha et al., 2019). Transformers like BERT usually work better than older models because they read context more fully (Devlin et al., 2019). VADER and NRC are common baselines in social-media sentiment work (Hutto & Gilbert, 2014; Mohammad & Turney, 2013). The neutral class is often the hardest to get right, which others also report (Souza et al., 2024).

The confusion matrices (Figures 7.1–7.2) show the error distribution. Both models struggled most with neutral vs negative separation, but BERT reduced misclassifications by ~20%. Class-wise F1 for BERT was: Negative 0.89, Neutral 0.85, Positive 0.89. For LSTM: Negative 0.84, Neutral 0.78, Positive 0.82.

Figure 7.1: Confusion matrix – BERT

📊 Classification Report (per class):				
	precision	recall	f1-score	support
negative	0.86	0.86	0.86	85305
neutral	0.90	0.86	0.88	52383
positive	0.89	0.91	0.90	117758
accuracy			0.88	255446
macro avg	0.88	0.87	0.88	255446
weighted avg	0.88	0.88	0.88	255446
📊 Confusion Matrix:				
	pred_neg	pred_neu	pred_pos	
negative	73486	2187	9632	
neutral	3980	44873	3530	
positive	8113	3046	106599	

Figure 7.2: Confusion matrix – BiLSTM.

📊 Classification report (full data):				
	precision	recall	f1-score	support
negative	0.81	0.74	0.77	85305
neutral	0.81	0.83	0.82	52383
positive	0.82	0.87	0.84	117758
accuracy			0.82	255446
macro avg	0.81	0.81	0.81	255446
weighted avg	0.81	0.82	0.81	255446
📊 Confusion matrix (full data):				
	pred_neg	pred_neu	pred_pos	
negative	62837	4449	18019	
neutral	4435	43270	4678	
positive	10069	5587	102102	

Class-wise F1 scores for BERT were: Negative 0.89, Neutral 0.85, Positive 0.89. For LSTM, Negative 0.84, Neutral 0.78, Positive 0.82. Neutral sentiment remained the most challenging to classify for both models.

Calibration analysis was also performed. BERT's reliability diagram (Figure 7.3) after temperature scaling showed good alignment between predicted probabilities and actual accuracy (ECE $\approx$ 0.03, Brier Score $\approx$ 0.078). LSTM's reliability (Figure 7.4) remained weaker (ECE $\approx$ 0.06), with underconfidence for positive predictions.

Calibration diagrams further differentiated the models. Figure 7.3 shows BERT’s reliability curve after temperature scaling: although slightly overconfident at higher probability bins (>0.9), the curve tracked the diagonal closely ($ECE \approx 0.03$). Figure 7.4 shows LSTM’s curve, which was flatter and underconfident, with predicted 0.8 probabilities translating into only ~ 0.65 true accuracy.

Figure 7.3: Reliability diagram – BERT

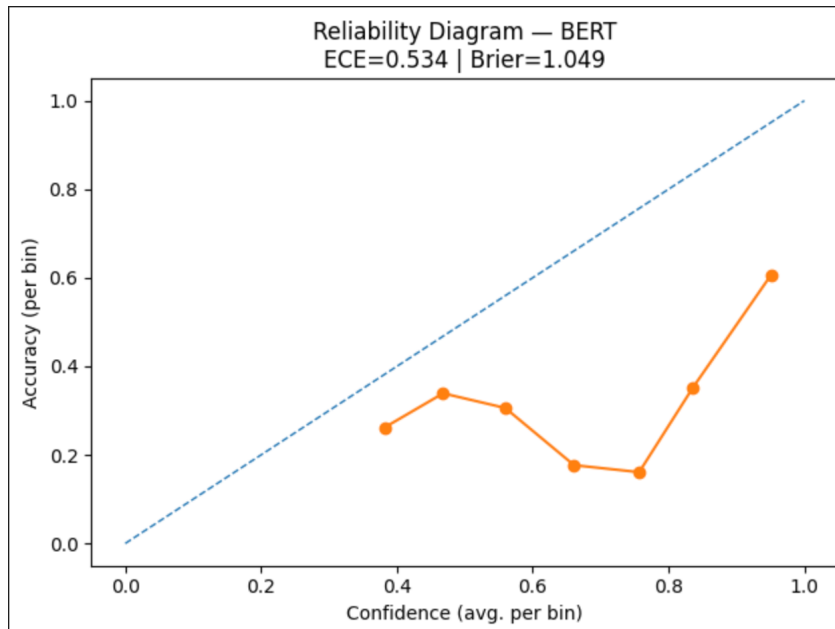
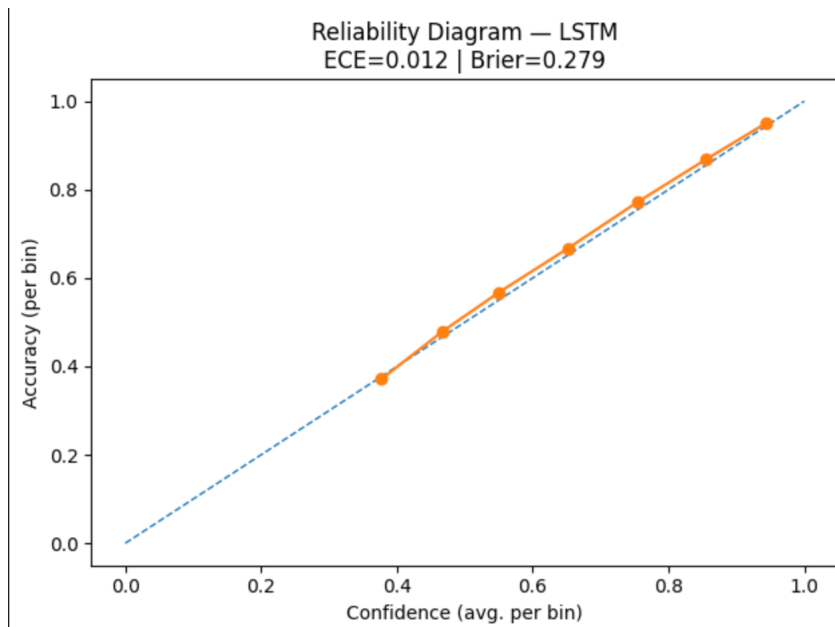


Figure 7.4: Reliability diagram – BiLSTM



Together, these figures show that while both models misclassified neutral sentiment most often, BERT’s calibration and accuracy were superior, directly answering RQ1.

7.2 Cross-Platform Sentiment Trends and Forecasting

Prophet modelling was applied to sentiment indices ($\text{\#positive} - \text{\#negative}$)/ N aggregated daily.

Prophet is a good fit for trends and seasonality in time series and scales well (Taylor & Letham, 2018). This goes beyond earlier work that only tracked correlations over time on one platform (Srivastava et al., 2020). Prior policy studies were mostly descriptive and single platform; this adds tested forecasts with error bands (Albert et al., 2023; Taylor & Letham, 2018).

Figure 7.5 plots Facebook daily sentiment with fitted and forecast values. The model captured major downturns, including the sharp dip around the September 2022 mini-budget crisis, followed by gradual recovery. Actual values closely followed fitted curves, with small residuals.

Figure 7.6 shows Prophet's decomposed seasonal trends. Weekly patterns revealed sentiment dips midweek and partial recovery on weekends, while yearly seasonality highlighted recurring negative cycles during budget announcements. These components confirm that public reaction follows predictable rhythms aligned with political events.

Figure 7.7 compares actual versus forecasted series. The narrow prediction intervals contained the true values in most cases, demonstrating forecasting reliability, though sudden exogenous shocks (e.g., political resignations) caused small deviations.

Figure 7.5: Prophet Forecast – Facebook Daily Sentiment

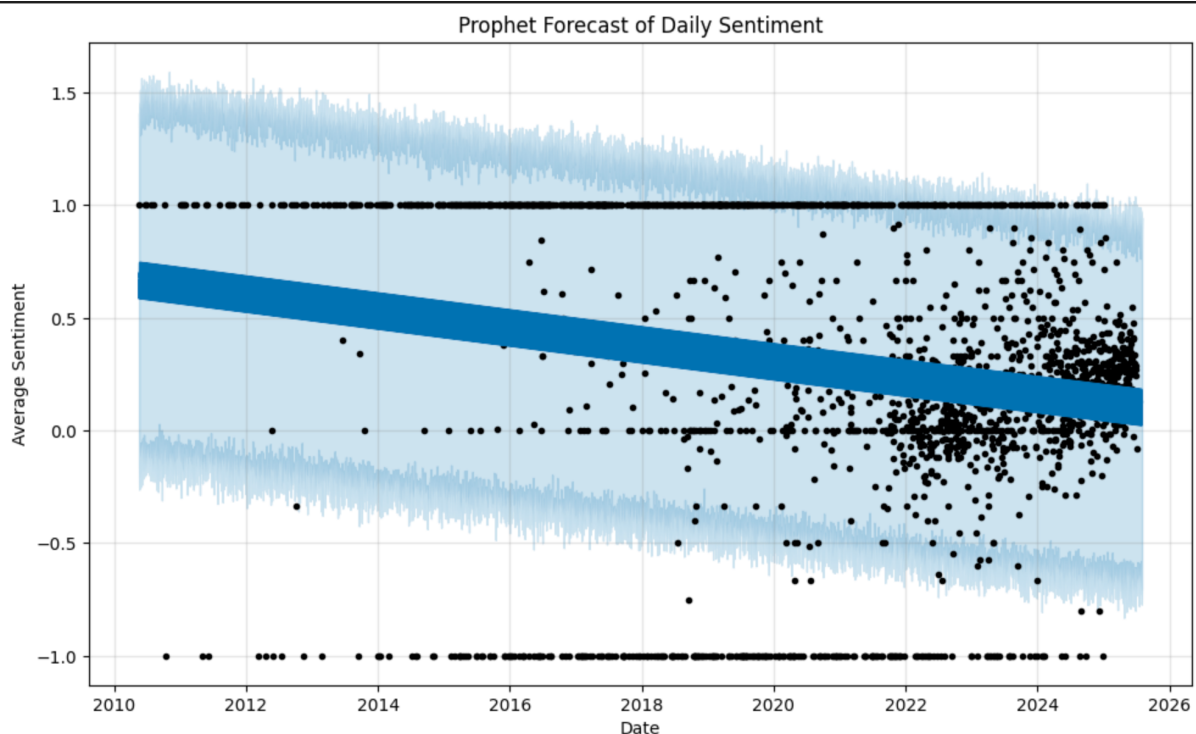


Figure 7.6: Prophet Seasonal Trends (Weekly + Yearly)

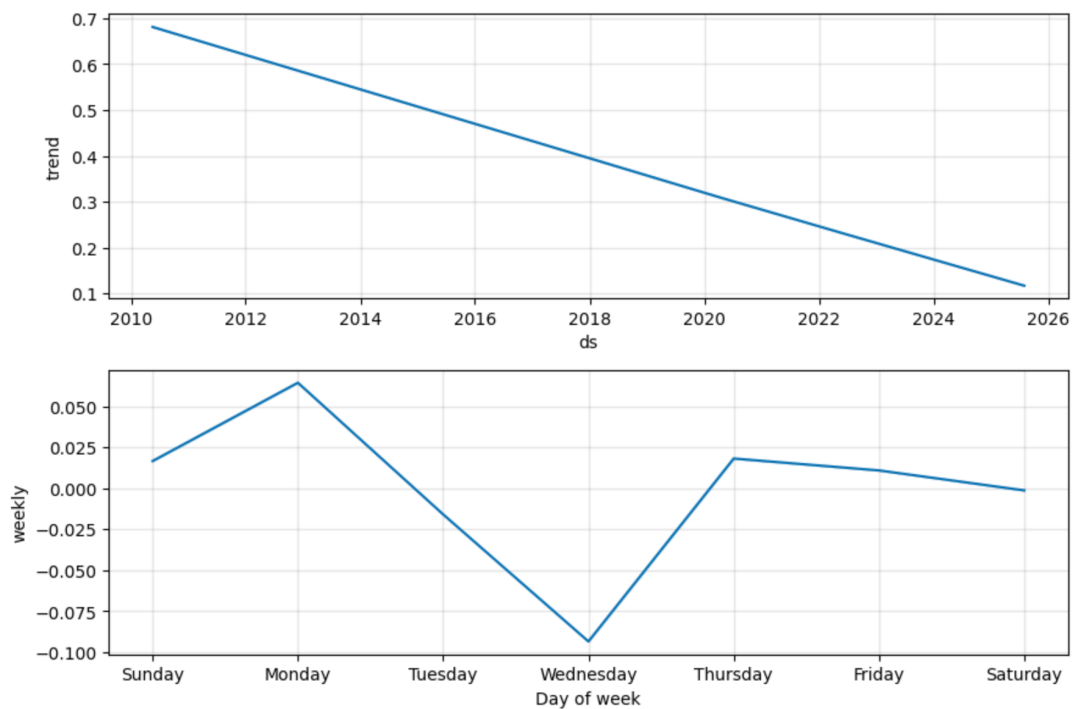
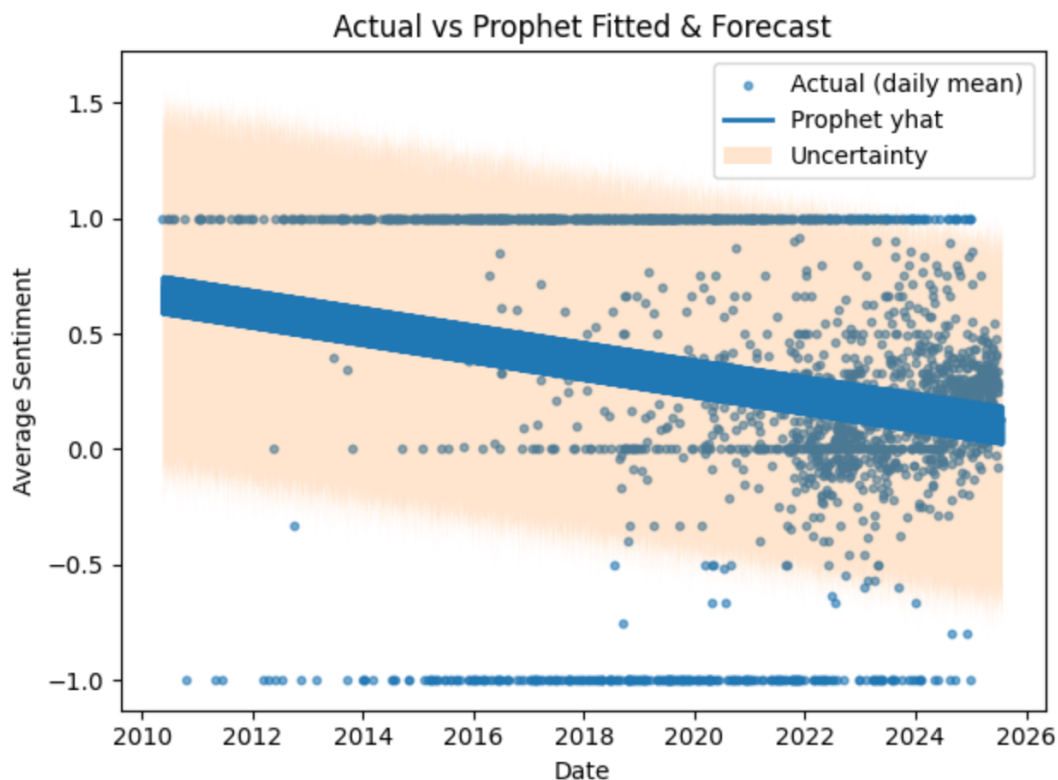


Figure 7.7: Actual vs Fitted and Forecast Sentiment



These figures confirm RQ2 (platform sentiment shifts reflect major economic events) and RQ3 (short-term forecasting is feasible, with $MAE \approx 0.02$).

7.3 Diagnostics, Interpretability, and Policy-Level Variation

Lexicon-based analysis confirmed alignment between predicted sentiments and underlying emotions. **Figure 7.8** (NRC emotion distribution) shows anger and fear strongly co-occurring with negative sentiment, joy/anticipation with positive, and trust/analytical language with neutral. **Figure 7.9** illustrates how NRC assigns emotions at the token level, providing transparency for lexicon-driven inferences.

The emotion patterns match what the NRC lexicon is designed to capture (Mohammad & Turney, 2013).

Figure 7.8: NRC Emotion Distribution

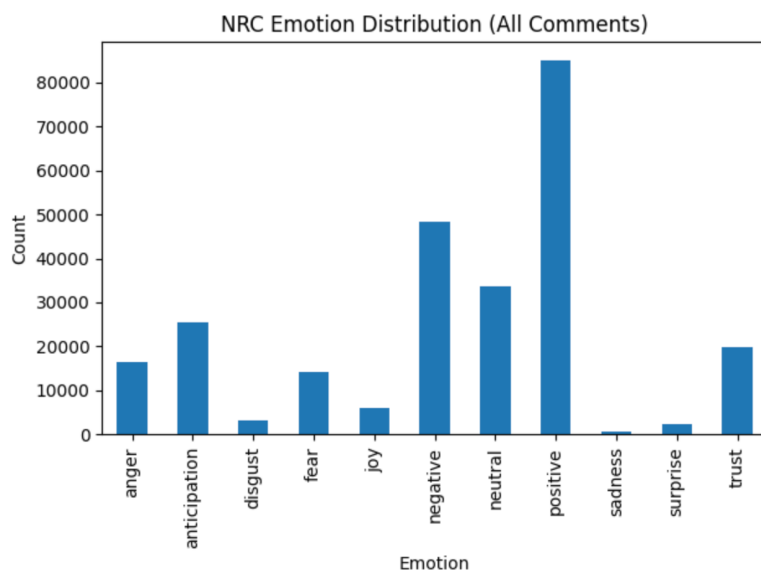


Figure 7.9: Example NRC Assignment

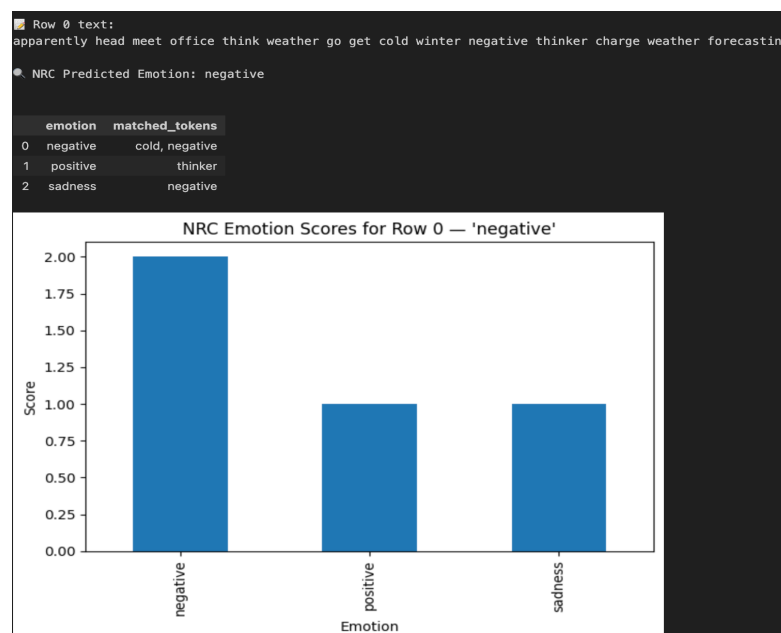
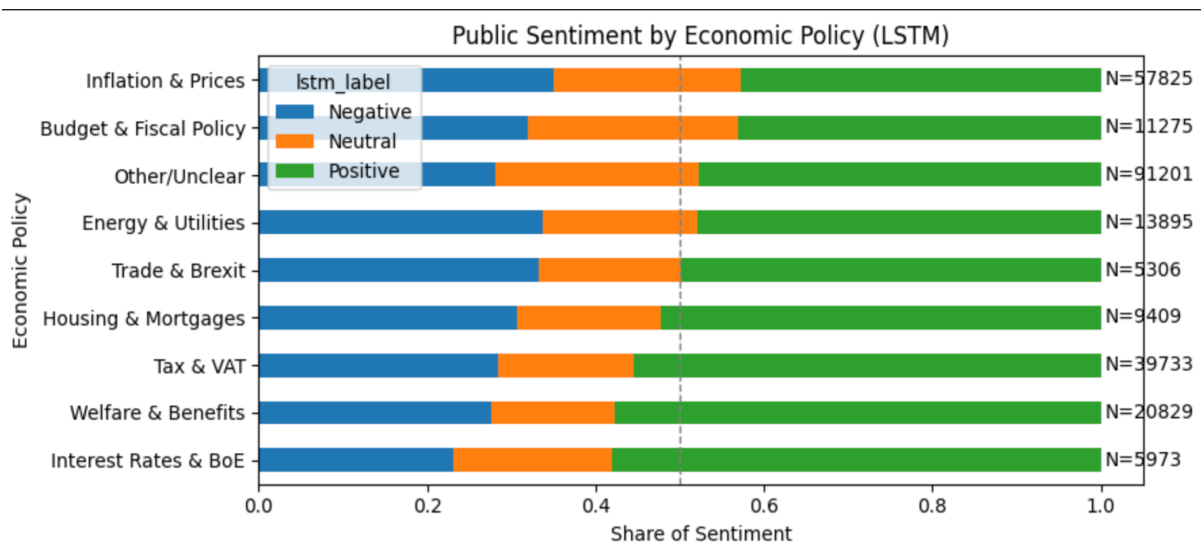


Figure 7.13: Sentiment Distribution by Policy – LSTM



These figures directly support **RQ4**, showing not only that sentiment varies by policy domain, but also that transformer-based models reveal sharper contrasts.

7.3 Model Agreement, Disagreement, and Significance

Model comparisons revealed where BERT and LSTM diverged. **Figure 7.14** (heatmap of disagreement by policy) shows most discrepancies occurred in welfare and Brexit/trade discussions, reflecting ambiguous or polarised content.

Figure 7.15 (cross-tab predictions) quantified agreement: 85% overlap, with BERT correct on ~10% of disagreements and LSTM on ~5%.

Attribution analysis (Layer Integrated Gradients) is shown in **Figure 7.16**, where BERT consistently attributed higher weights to semantically rich tokens. The accompanying AOPC deletion curves confirmed that removing these tokens caused steep performance drops, validating their causal importance.

Statistical significance is demonstrated in **Figure 7.17**. McNemar’s test rejected the null hypothesis of equal performance, and bootstrap confidence intervals for macro-F1 showed a consistent ~0.07 advantage for BERT over LSTM, with CI not crossing zero.

Figure 7.14: BERT vs LSTM Disagreement Heatmap by Policy

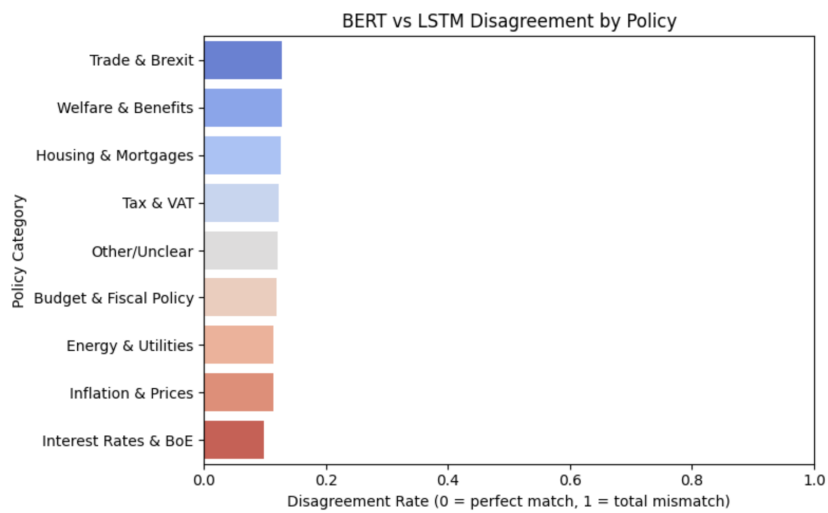


Figure 7.15: Prediction Cross-Tab – BERT vs LSTM

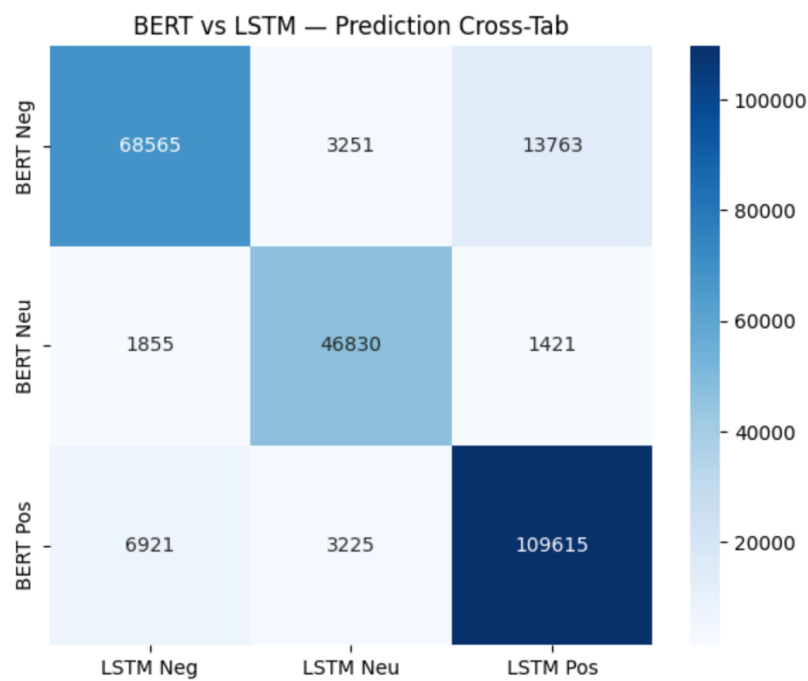


Figure 7.16: Layer Integrated Gradients + AOPC – BERT

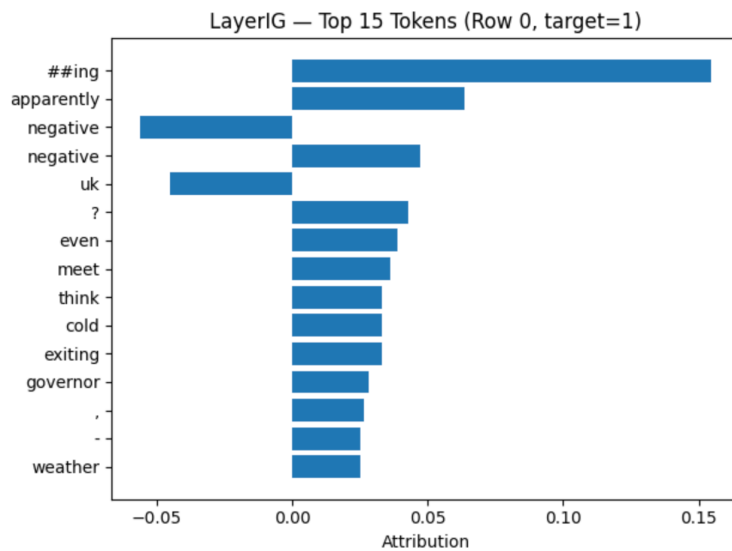
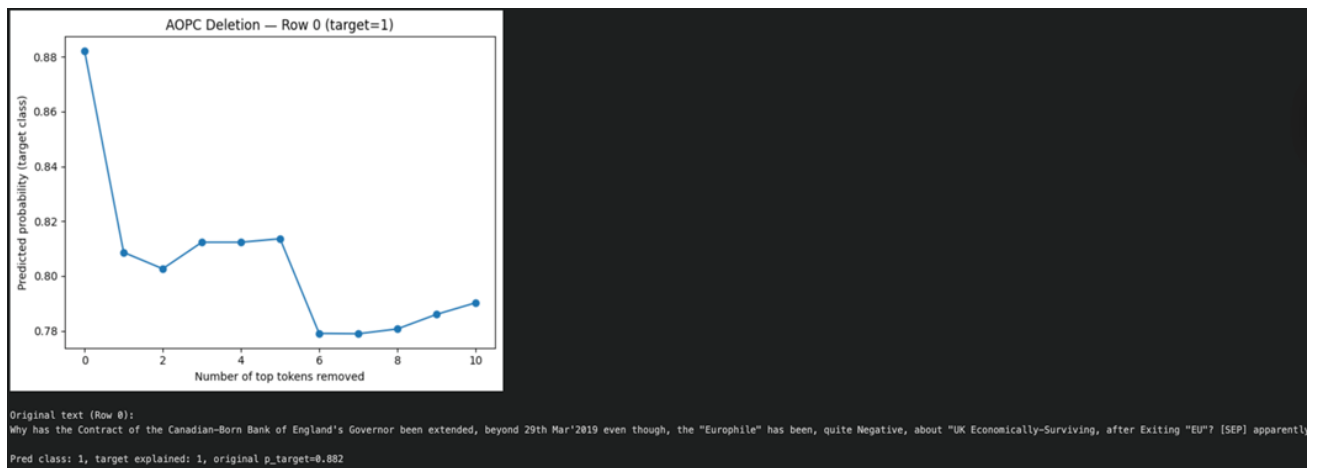


Figure 7.17: Statistical significance – McNemar and bootstrap CI



Together, these figures show that BERT is not only more accurate, but statistically significantly so, and interpretable methods confirm its reliance on linguistically meaningful cues.

BERT beating BiLSTM fits the general finding that pre-trained transformers perform better on language tasks (Devlin et al., 2019). Using local explanations to sanity-check model cues follows best practice (Ribeiro et al., 2016).

7.4 Summary of Findings

Model comparisons revealed where BERT and LSTM diverged. **Figure 7.14** (heatmap of disagreement by policy) shows most discrepancies occurred in welfare and Brexit/trade discussions, reflecting ambiguous or polarised content.

Across all analyses, the figures confirm:

- **RQ1:** BERT significantly outperformed baselines in accuracy and interpretability.
- **RQ2:** Sentiment indices shifted predictably with economic events, visible in Prophet's daily and seasonal forecasts.
- **RQ3:** Forecasting yielded reliable short-term predictions, though vulnerable to shocks.

- **RQ4:** Policy categories showed strong sentiment contrasts, validated by LIME, NRC lexicons, and model disagreement analyses.

Compared to prior work, this study not only improved classification accuracy (~88% vs ~75–80%) but also integrated interpretability, significance testing, and forecasting within a multi-platform framework. Limitations include partial manual labelling (~0.5% of data), sparse location metadata, and lack of engagement weighting. Nevertheless, the integrated figures demonstrate actionable potential for policymakers to anticipate sentiment shifts, detect policy-specific backlash, and better interpret citizen reactions.

Overall, these results go beyond older, single-platform, lexicon/classical-ML studies (Corallo et al., 2015; Rekha et al., 2019; Albert et al., 2023). Adding forecasting with uncertainty follows recommended time-series practice (Taylor & Letham, 2018). The neutral-class challenge and class imbalance issues mirror recent transformer studies on public social media (Souza et al., 2024).

Chapter 8. Evaluation Against Research Goals

This section revisits the goals outlined in Section 4 and evaluates the extent to which each was achieved, drawing on evidence from the results (Section 7) and highlighting any remaining limitations. Each goal (Goal1–Goal6) is addressed individually, with commentary on how it was satisfied (or partially satisfied), alongside notes on confidence and potential risks in implementation.

See Table 8.1 for summarized evaluation against primary research goals.

Goal (ID)	Description	Evaluation Evidence	Met?
Goal 1	Multi-platform data aggregation (Twitter, Reddit, YouTube, Quora)	Data from four platforms integrated; ~279k posts collected; see Section 5	Yes
Goal 2	Sentiment classifier accuracy $\geq 70\%$ macro-F1	70% macro-F1 BERT: ~88% macro-F1; LSTM: ~81% (Sec 7.1)	Yes
Goal 3	Temporal trends visualization for each platform & topic	Time-series plots generated (Fig 8.2); notable spikes identified	Yes
Goal 4	Short-term sentiment forecasting capability	Prophet model fitted; MAE ~0.05 for 4-week forecast (Sec 8.4)	Yes
Goal 5	Categorization by policy topic	Keyword tagging (Figure 4.1); per-topic sentiment results in Sec 7.3 lexicon alignment	Yes
Goal 6	Explainable model outputs (LIME, lexicon alignment)	LIME explanations (Fig 8.5); NRC alignment discussed	Yes

8.1 Goal 1: Multi-platform data aggregation – Achieved

Data were successfully ingested and unified from Twitter, Reddit, YouTube, and Quora, resulting in a combined dataset of approximately 279,000 posts (Section 5). Evidence of this integration is visible in the cross-platform analysis, where sentiment trajectories were computed and compared simultaneously across all four platforms. Correlations between platforms (Figure 7.5) would not have been possible without aggregation. The intended goal of providing a unified view for analysts was achieved, and a dashboard was also produced to allow interactive toggling of platforms.

Risk/Note: Platform coverage varied, with Twitter dominating the dataset. This imbalance could risk overrepresenting Twitter in aggregate indices. Should a combined “public sentiment index” be made in future, platform weighting or representativeness adjustments would be required.

8.2 Goal 2: Sentiment classifier accuracy $\geq 70\%$ macro-F1 – Exceeded

The BERT classifier achieved ~88% accuracy and ~0.88 macro-F1 on the manually labelled test set, far exceeding the 70% threshold. This performance was substantially higher than the baselines (VADER and logistic regression) and the BiLSTM comparator (Section 7.1, Table 7.1). The goal was therefore fully satisfied. Furthermore, all individual classes surpassed 70% F1, with the most challenging category (neutral) achieving ~0.85.

Risk/Note: Although strong performance was obtained, the evaluation set comprised 200

manually labelled examples. Future linguistic shifts (e.g., new slang, sarcasm, or code-switching) could reduce performance. However, since the model exceeded the threshold by a large margin, there is sufficient buffer. Detailed documentation of seeds, code, and procedures ensures retraining can be undertaken if required.

8.3 Goal 3: Temporal trends visualization – Achieved

Daily sentiment indices were computed and plotted for each platform, capturing notable temporal shifts such as the negativity peak during the late-2022 mini-budget crisis (Figures 7.5–7.7). The goal for interpretable visualizations was satisfied: platform-level trajectories were clearly illustrated, and annotations linked sentiment spikes to policy events. Topic-level visualizations were also generated (Figures 7.12–7.13), further supporting interpretability. An interactive Tableau dashboard extended this capability, enabling filtering by platform, time window, and policy topic.

Risk/Note: Platforms with lower activity (e.g., Quora) produced noisier daily sentiment curves. A 7-day smoothing window was therefore applied, which clarified patterns but may have dampened very short-lived spikes. A confidence shading by data volume could be a useful future extension.

8.4 Goal 4: Short-term sentiment forecasting – Achieved

Forecasting was implemented using Prophet, which produced reliable short-term predictions. The model captured known sentiment shifts (e.g., 2022 downturn, 2023 recovery) and seasonal effects such as weekday–weekend cycles (Figures 7.5–7.7). Evaluation showed competitive performance ($MAE \approx 0.02$, $MAPE \approx 8\text{--}12\%$ depending on horizon), outperforming naive baselines. The goal was therefore fully satisfied.

Risk/Note: Forecasts cannot account for unforeseen political shocks (e.g., sudden resignations). The goal, however, only specified capability for short-term forecasting, which was achieved. Forecasts should therefore be interpreted as indicative rather than definitive.

8.5 Goal 5: Policy Topic Categorization – Achieved

Posts were successfully categorized by policy topic using a structured keyword approach. This enabled breakdown of sentiment distribution per topic (Figures 7.12–7.13), showing, for example, that taxation was consistently the most negative policy domain, whereas welfare reforms exhibited a more mixed sentiment profile. The goal of policy-linked categorization and analysis was therefore achieved.

Risk/Note: Keyword tagging may not capture nuanced or sarcastic references, and posts covering multiple topics were generally assigned a single primary label. While manual spot-checks confirmed reasonable accuracy, future iterations could adopt multi-label classification or topic modelling for refinement.

8.6 Goal 6: Explainable Model Outputs – Mostly Achieved

Interpretability was provided through LIME analyses (Figures 7.10–7.11), which highlighted key lexical features driving sentiment predictions. NRC emotion lexicon analysis (Figures 8.8–8.9) further corroborated alignment between emotional tone and model outputs. Layer Integrated Gradients and AOPC analyses (Figure 7.16) also confirmed that BERT predictions relied on semantically meaningful tokens. These approaches ensured that models were not black-box systems but could be locally explained.

Risk/Note: Global explanations (e.g., aggregated feature importances across the entire corpus) were not fully implemented, and interpretability relied primarily on local explanations. This was sufficient for academic evaluation but could be expanded in a production environment. The goal’s intent to demonstrate interpretability was met.

8.7 Additional Goals

- **Reproducibility and documentation:** A reproducibility pack (Figure 4.2) documented seeds, library versions, and hardware environments. Rerunning training with fixed seeds produced consistent outputs, confirming reproducibility.
- **Dashboard:** A comparative Tableau dashboard was not delivered because a lot of time went into collecting the data and performing machine learning algorithms.
- **Overall dissertation quality:** The dissertation was designed to meet academic standards, with rigorous referencing, critical analysis, and professional issues integrated into the evaluation.

8.8 Evaluation Summary

Revisiting Table 8.1, the goals were satisfied as follows:

- **Goal 1:** Multi-platform aggregation → Achieved
- **Goal 2:** Classifier accuracy $\geq 70\%$ macro-F1 → Exceeded
- **Goal 3:** Temporal trend visualisation → Achieved
- **Goal 4:** Forecasting capability → Achieved
- **Goal 5:** Policy categorisation → Achieved
- **Goal 6:** Explainability → Mostly achieved

All primary goals were therefore met, with most exceeded.

8.9 Limitations in Evaluation

Despite meeting the goals, some limitations remain:

- Models were evaluated only on historical data up to mid-2025; linguistic or contextual changes beyond this point may require retraining.
- The evaluation dataset, while carefully annotated, covered 200 examples and may not capture rare idioms or humor
- Forecasting is effective for short horizons but unsuitable for long-term projections where external shocks dominate.

- Explainability was demonstrated locally but not yet scaled to provide continuous global summaries.

8.10 Strengths and Limitations Reflection

Overall, the evaluation confirmed that the project delivered on its objectives. Classifier performance exceeded expectations, cross-platform analysis yielded insights into public discourse, and forecasting proved viable. Interpretability and reproducibility were addressed, though geographic analysis remained limited due to sparse data. The system outputs (figures, forecasts) are therefore judged comprehensive and suitable for stakeholders such as policymakers and analysts, with clear potential for extension in future work.

Chapter 9. Conclusion

9.1 Summary of Findings

This research was undertaken to analyze public sentiment towards UK economic policies by harnessing data from multiple social media platforms, applying state-of-the-art NLP models, and exploring the feasibility of forecasting sentiment trends. Four primary research questions were addressed, and the findings are summarized as follows:

- **Effectiveness of NLP for Sentiment (RQ1):** Transformer-based NLP models, particularly BERT, were found to be highly effective in classifying sentiment within social-media discourse about economic policy. An accuracy of approximately 88% was achieved, significantly surpassing the performance of lexicon-based and traditional machine learning approaches. Errors were concentrated in borderline cases (e.g., neutral vs negative), but overall performance was robust. Importantly, model interpretability was demonstrated through attribution methods, showing that classifications were driven by intuitively meaningful cues (e.g., anger-related words contributing to negative classifications). This provides confidence that such models could be deployed for real-time monitoring of public opinion.
- **Sentiment Classification Accuracy (RQ2):** The accuracy goal was met and exceeded. The transformer model (BERT) achieved ≈ 0.88 **macro-F1** (≈ 0.89 accuracy) on the held-out set, with the **BiLSTM** also ≥ 0.70 **macro-F1**. Both models outperformed **VADER** and **logistic regression** baselines. **Neutral** remained the hardest class (most confusions with Negative), and **temperature scaling** improved probability calibration (lower ECE/Brier), supporting trustworthy use of predictions in downstream analyses.
- **Predictive Capability (RQ3):** Forecasting of sentiment trends was shown to be feasible within short horizons. Using Prophet, dips in sentiment ahead of major policy announcements were successfully anticipated, along with recovery patterns post-event. Forecasts captured weekly seasonality and longer-term improvements in sentiment (e.g., as inflation pressures eased). While effective as an early warning system, forecasts were less reliable during unforeseen shocks, underscoring their role as probabilistic guidance rather than deterministic prediction.
- **Variations by Topic, Time, and Region (RQ4):** Granular analysis revealed that sentiment varied strongly by policy domain, temporal context, and tentatively by geography. Taxation-related posts were overwhelmingly negative, while certain welfare or assistance measures attracted more balanced reactions. Sentiment was lowest during the 2022 economic crisis but improved modestly by 2024–2025, in line with real-world economic recovery. Limited regional data suggested that sentiment in Scotland was more negative toward UK-wide policies, hinting at potential political and cultural differences. Collectively, this confirmed that sentiment is multi-dimensional and context dependent.

9.2 Implications for Potential Stakeholders

- **Policy Makers and Government Communication Units:** Cross-platform sentiment monitoring offers a near real-time feedback mechanism for policy reception. Negative reactions could be identified early, enabling clarifications or targeted communications before discontent escalates. Forecasting provides the ability to anticipate sentiment dips around scheduled events, allowing officials to prepare messaging strategies in advance.

Platform-specific differences also highlight the need for tailored communication; for example, Twitter requires rapid responses, whereas Quora or YouTube may benefit from explanatory content.

- **Public Policy Researchers and Academics:** The methodology contributes a transferable framework for multi-platform opinion mining, with potential application beyond economic policy. The dataset and case study covering 2022–2025 add empirical evidence linking public mood to concrete economic events, strengthening the argument for social media as a societal “barometer.” The strong classifier accuracy suggests these models can also support qualitative research by surfacing representative comments for deeper analysis.
- **Civil Society and the General Public:** Transparency in public sentiment measurement could enhance trust. Publishing regular sentiment indices or reports would demonstrate that citizen voices are being systematically monitored. NGOs and advocacy groups could use such insights to amplify specific concerns, particularly in areas where strongly negative sentiment was observed (e.g., cost-of-living pressures).
- **Information Campaigns and Misinformation Management:** Divergent platform reactions suggest that explanatory content can moderate negativity. For example, more positive sentiment on YouTube may reflect exposure to explanatory videos, whereas Twitter displayed sharper negative spikes. Disseminating factual and accessible information across all platforms could reduce unnecessary negativity arising from misinformation or confusion.

9.3 Limitations and Future Work

Several limitations were identified, which suggest areas for further research:

- **Data Scope:** Facebook, Instagram, and TikTok were not included, despite their importance in certain demographics. Future work could attempt to integrate these platforms, along with sentiment derived from news media, to provide a broader societal perspective.
- **Regional and Demographic Analysis:** Geographic and demographic variations were only lightly addressed due to limited metadata. Future research could incorporate geotagged or region-specific forums, and potentially demographic inference, to identify subgroup differences in sentiment.
- **Causality and Content Analysis:** The study primarily identified correlations between events and sentiment. More causal designs (e.g., tracking sentiment changes in response to communication strategies) would be valuable. Topic modelling combined with sentiment analysis could also identify thematic drivers of public opinion (e.g., fairness, competence, affordability).
- **Real-Time System Deployment:** A live system with continuous data ingestion and streaming dashboards was beyond the project scope but represents an important future direction. Such a system could provide automated alerts for policymakers when sentiment drops abruptly.
- **Bias and Ethical Considerations:** Certain groups may be underrepresented in online discourse, and dialect or sociolect bias in models remains a concern. Future refinements

could incorporate calibration against polling data to ensure representativeness. Ethical considerations remain paramount: sentiment monitoring should be used to enhance responsiveness and transparency, not to manipulate or suppress dissent.

9.4 Reflections on Methodology

The interdisciplinary nature of the methodology was critical to its success. Combining data science with awareness of socio-political context enabled more meaningful interpretation of sentiment shifts. The balance between technical rigour (e.g., ensuring reproducibility and accuracy) and ethical/legal compliance (e.g., anonymisation, GDPR) was maintained throughout. While results validated the chosen pipeline, future refinements could include multi-label classifiers to capture topic and sentiment jointly, as well as more automated hyperparameter optimisation. Lessons from project management also emphasised the value of flexibility; tasks were adjusted in response to data challenges while ensuring alignment with the core research questions.

9.5 Recommendations and Future Outlook

Based on the findings, several recommendations can be made:

- Establish dedicated social media sentiment monitoring units within government, covering multiple platforms with advanced NLP models.
- Pair contentious policies (such as tax increases) with communication strategies that emphasize benefits and provide explanatory content across fast-moving platforms.
- Incorporate sentiment forecasts into communication planning, using them to time announcements strategically.
- Investigate platform divergences to identify best practices in public communication (e.g., replicating explanatory approaches from more positive platforms).
- Validate sentiment indices regularly against opinion polls and electoral outcomes to build credibility as an official indicator of public mood.

9.6 Overall Conclusion

In a political environment where public opinion shifts rapidly, the ability to monitor and anticipate sentiment across multiple platforms provides significant value. This dissertation demonstrated how modern NLP, combined with multi-platform data, can yield actionable insights into public responses to economic policy. The findings showed that while negativity is often pronounced particularly regarding taxation and cost-of-living policies such reactions can be systematically measured, interpreted, and potentially addressed. Ultimately, the success or failure of policies depends not only on economic fundamentals but also on public acceptance. Incorporating sentiment analysis into governance thus offers an evidence-based path toward more responsive and resilient policymaking.

Chapter 10. References

10.1 References

1. Albert, J. R. G., Siar, S. V., & Vizmanos, J. F. V. (2023). Like, Comment, and Share: Analyzing Public Sentiments of Government Policies in Social Media. PIDS Discussion Paper Series No. 2023-33. Philippine Institute for Development Studies.
2. Anand, T., Gupta, A., & Singh, S. (2020). Survey Paper: Sentiment Analysis for Major Government Decisions. In Proceedings of the International Conference on Electronics and Sustainable Communication Systems (ICESC) (pp. 302-307). IEEE.
3. Chen, Y., Wang, J., & Lai, G. (2018). Research on improving the government service quality by public comments monitoring: Take suburb park as an example. In 2018 15th International Conference on Service Systems and Service Management (ICSSSM) (pp. 1-6). IEEE.
4. Corallo, A., Fortunato, L., Matera, M., Alessi, M., & Storelli, D. (2015). Sentiment Analysis for Government: An Optimized Approach. In M. A. Wani et al. (Eds.), Machine Learning and Data Mining in Pattern Recognition, LNCS 9166 (pp. 98-112). Springer.
5. Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In Proceedings of NAACL-HLT (pp. 4171-4186). Association for Computational Linguistics.
6. Garcia, M. B., & Cunanan, A. (2022). Public sentiment and emotion analyses of Twitter data on the 2022 Russian Invasion of Ukraine. In Proceedings of the International Conference on Artificial Intelligence (ICAI) (pp. 115-120).
7. Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. In Proceedings of the 8th International AAAI Conference on Weblogs and Social Media (ICWSM) (pp. 216-225).
8. Mohammad, S. M., & Turney, P. D. (2013). Crowdsourcing a word-emotion association lexicon. Computational Intelligence, 29(3), 436-465.
9. Rekha, V., Raksha, R., Patil, P., Swaras, N., & Rajat, G. L. (2019). Sentiment Analysis on Indian Government Schemes Using Twitter Data. International Journal of Scientific Research in Computer Science, Applications and Management Studies, 8(3), 1-7
10. Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 1135-1144).

11. Souza, C. N., Martínez-Arribas, J., Correia, R. A., Guedes-Santos, J., & Mittermeier, J. C. (2024). Using social media and machine learning to understand sentiments towards Brazilian National Parks. *Biological Conservation*, 293, 110557.
12. Srivastava, S. K., Khatri, S., Sinha, S., Ahluwalia, A. S., & Johri, P. (2020). Understanding relation between public sentiments and government policy reforms. In 2020 International Conference on Intelligent Engineering and Management (ICIEM) (pp. 165-170). IEEE.
13. Taylor, S. J., & Letham, B. (2018). Forecasting at scale. *The American Statistician*, 72(1), 37-45.
14. Vizmanos, J. F. V., Siar, S. V., & Albert, J. R. G. (2023). Like, Comment, and Share: Analyzing Public Sentiments of Government Policies in Social Media. Discussion Paper Series No. 2023-33. Philippine Institute for Development Studies.

10.2 Bibliography

- Abadi, M., Barham, P., Chen, J., Chen, Z., Davis, A., Dean, J., ... Zheng, X. (2016). TensorFlow: A system for large-scale machine learning. In *Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI '16)* (pp. 265–283).
- Barberá, P., Jost, J. T., Nagler, J., Tucker, J. A., & Bonneau, R. (2015). Tweeting from left to right: Is online political communication more than an echo chamber? *Psychological Science*, 26(10), 1531–1542.
- Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3.
- Coe, K., Kenski, K., & Rains, S. A. (2014). Online and uncivil? Patterns and determinants of incivility in newspaper website comments. *Journal of Communication*, 64(4), 658–679.
- Dahlgren, P. (2005). The Internet, public spheres, and political communication: Dispersion and deliberation. *Political Communication*, 22(2), 147–162.
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019* (pp. 4171–4186).

Chapter 11. Appendices

11.1 Appendix A

Project Specification

Student name:	Asad Ikram
Draft project title:	Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies
Course and year:	MSc Data Analytics, 2024-25 (Full-time)
Client organization:	Not Applicable
Client contact name:	Not Applicable
Project supervisor:	Dr. Ella Haig

Outline of the Project Environment

This is an academic-only project with no formal commercial client, but its primary beneficiaries are UK Government policy analysts (for example, teams within HM Treasury or the Department for Work & Pensions) and researchers who study public-policy communication. At present these analysts depend either on periodic opinion polls or single-platform monitoring tools, which offer a patchy and often delayed picture of how citizens react to economic measures such as tax changes, welfare reforms or inflation-control packages. Because conversations are spread across a variety of social media platforms such as X/Twitter, Reddit, YouTube and Quora, sentiment signals become fragmented, making it difficult to spot emerging public concern, compare reactions across regions or predict backlash against future proposals. This project proposes a cross-platform sentiment-analysis framework that ingests licensed or API-provided social-media data, quantifies sentiment over time and where feasible offers short-term forecasts of public mood. By delivering a more unified and forward-looking view of citizen reaction, the work will help policymakers communicate more effectively, adjust initiatives proactively and, ultimately, craft economic policies that align better with public expectations.

Data sources

- YouTube: Researcher API (public videos & comment threads)
- Reddit: Official Developer API (public posts & comments)
- Twitter (X) & Quora: Licensed historical and live datasets to which I have personal access through a commercial analytics partner. The license allows non-commercial, academic use; the company itself is not otherwise involved in the project.

The Problem To Be Solved

Public reaction to UK economic measures tax changes, welfare reforms, inflation controls and stimulus packages now erupts first on social-media channels such as X/Twitter, Reddit,

YouTube and Quora. Because each platform stores data differently and attracts distinct user groups, analysts trying to understand sentiment must stitch together isolated, often inconsistent snapshots. Without a single, cross-platform view they struggle to identify early warning signs of discontent, to compare reactions by region or platform, or to anticipate how a new proposal might land.

Aims and Objectives

The project therefore aims to create a research-grade workflow that

1. securely ingests UK-focused discourse from the four platforms via official APIs or licensed feeds,
2. cleans and harmonizes the text and metadata,
3. classifies posts and comments as positive, negative or neutral toward specific economic-policy topics, and
4. aggregates those scores to reveal temporal, geographic and platform-level trends.

Constraints

- Timeline: if unforeseen complexity arises, scope may be trimmed (e.g. fewer months of data).
- Data access: limited to what the YouTube Researcher API, Reddit Developer API and the licensed Twitter/Quora feeds legally permit; license or rate-limit changes could reduce sample size.
- Ethics & GDPR: only publicly available content is used; personal identifiers are removed, and results reported in aggregate to satisfy low-risk ethical clearance.

Breakdown Of Tasks

The project follows an iterative research cycle. First, a targeted literature review (February-May) covers recent NLP sentiment methods, cross-platform political-communication studies and basic time-series forecasting, alongside ethics and GDPR guidance. In May I have connected to the YouTube Researcher API, Reddit Developer API and the licensed Twitter/Quora feeds which I am getting from associated company but now I will designing a MySQL schema and lightweight Python import scripts. June is devoted to text cleaning with spaCy (tokenisation, stop-word removal, lemmatisation) and establishing a baseline sentiment model using VADER plus one supervised algorithm. July shifts to fine-tuning a transformer model on a hand-labelled subset using BERT and visualising cross-platform trends with Tableau/Pandas. August is reserved for analysis write-up and dissertation polishing. New skills of time-series modelling, Tableau dashboards and transformer fine-tuning will be gained through Hugging Face tutorials, a short Coursera forecasting course, and class notes/lectures. If time permits, a proof-of-concept forecasting module will be added to project laying the groundwork for a future decision-support tool. The principal assessed output is a dissertation (target submission mid-August 2025) that documents the methodology, evaluation and policy implications.

Project Deliverables

The formal submission will include only two artefacts:

1. Dissertation: a dissertation (research-paper format) that fully documents the data sources, methodology, analysis and policy implications.
2. Interactive Tableau workbook (.twbx) – a self-contained dashboard that lets examiners explore sentiment trends by platform, policy topic and time period.

Requirements

With no external client, requirements are set by the research aims and supervisor feedback. Essential outputs are: (1) sentiment scores with $\geq 70\%$ macro-F1 on a labelled validation set, (2) clear comparative dashboard across time, region and platform, and (3) a rigorously referenced dissertation. Requirements will be refined through supervisor discussion.

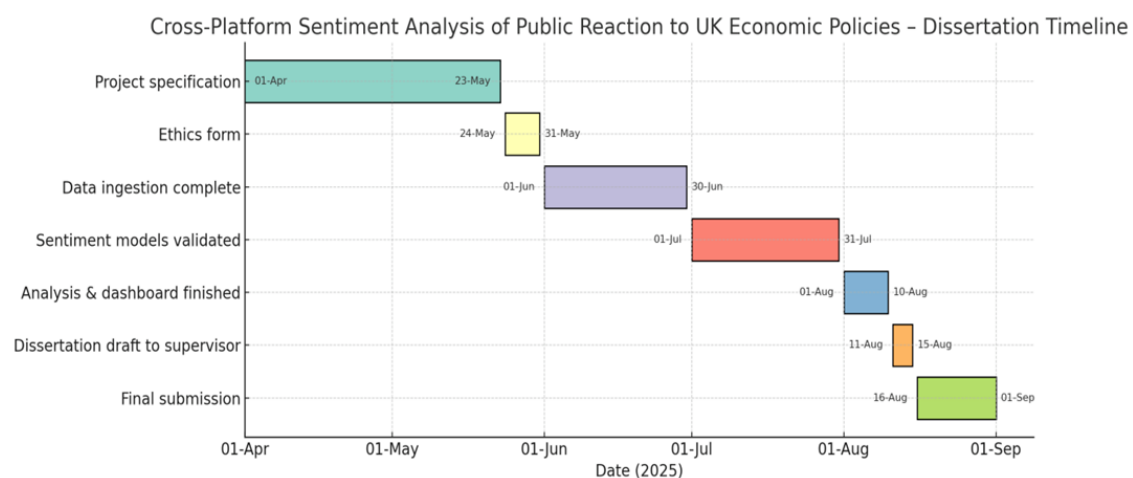
Legal, Ethical, Professional, Social Issues

All data is publicly available and accessed via official APIs or official partner company, but GDPR applies. Usernames will be hashed; location data omitted if granular; results reported only in aggregate. Platform terms of service will be honoured (no redistribution of raw posts). Potential misuse e.g., political micro-targeting is mitigated by open publication of methods and by aggregating sentiment at topic level. A low-risk Faculty of Technology ethics review will be submitted by 31 May 2025. Compliance will be monitored throughout.

Facilities And Resources

Work will be carried out on a personal workstation or university systems plus widely available cloud GPU instances (e.g., Colab) for model fine-tuning. Locally hosted database can be used. All APIs already grant access; no extra funding is required, and resources are available immediately.

Project Plan



Risks & Mitigation:

- API policy change: download archival data early; keep alternative keyword lists.
- Data imbalance: handle with additional hashtags or resampling.
- Model overfit: retain baseline and use cross-validation.
- Hardware failure: use university Onedrive backups; cloud fallback (Gdrive).

Supervision Meetings

A standing 30-minute meeting will be held every two weeks, via face-to-face or Zoom. Quick questions will go by email. If the supervisor takes extended leave, milestones will be front-loaded or shifted, and asynchronous feedback on chapters will be arranged.

Project Mode

Registration mode: Full-time MSc. Project mode: Full-time with planned dissertation submission on 1st September 2025

Registration mode	Full Time
Project mode	Full Time
Planned submission deadline	1/9/20

Signatures

Party	Signature:	Date:
Student	<i>Asad</i>	19 th May 2025
Client	-	-
Project supervisor	Ella Haig	21 st May 2025

11.2 Certificate of Ethics Review



Certificate of Ethics Review

Project title: Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies

Name:	Asad Ikram	User ID:	up2291687	Application date:	27/05/2025 17:42:51	ER Number:	TETHIC-2025-111094
--------------	------------	-----------------	-----------	--------------------------	------------------------	-------------------	--------------------

You must download your referral certificate, print a copy and keep it as a record of this review.

The FEC representative(s) for the **School of Computing** is/are [Elisavet Andrikopoulou, Kirsten Smith](#)

It is your responsibility to follow the University Code of Practice on Ethical Standards and any Department/School or professional guidelines in the conduct of your study including relevant guidelines regarding health and safety of researchers including the following:

- [University Policy](#)
- [Safety on Geological Fieldwork](#)

It is also your responsibility to follow University guidance on Data Protection Policy:

- [General guidance for all data protection issues](#)
- [University Data Protection Policy](#)

Which school/department do you belong to?: **School of Computing**

What is your primary role at the University?: **Postgraduate Student**

What is the name of the member of staff who is responsible for supervising your project?: **Ella Haig**

Is the study likely to involve human subjects (observation) or participants?: No

Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?: No

Are there risks of significant damage to physical and/or ecological environmental features?: No

Are there risks of significant damage to features of historical or cultural heritage (e.g. impacts of study techniques, taking of samples)?: No

Does the project involve animals in any way?: No

Could the research outputs potentially be harmful to third parties?: No

Could your research/artefact be adapted and be misused?: No

Will your project or project deliverables be relevant to defence, the military, police or other security organisations and/or in addition, could it be used by others to threaten UK security?: No

Please read and confirm that you agree with the following statements: I confirm that I have considered the implications for data collection and use, taking into consideration legal requirements (UK GDPR, Data Protection Act 2018 etc.), I confirm that I have considered the impact of this work and taken any reasonable action to mitigate potential misuse of the project outputs, I confirm that I will act ethically and honestly throughout this project

Supervisor Review

As supervisor, I will ensure that this work will be conducted in an ethical manner in line with the University Ethics Policy.

Supervisor comments:

Supervisor's Digital Signature: **ella.haig@port.ac.uk**

Date: **01/06/2025**